Machine learning

Neural networks/deep learning

Thomas D. Nielsen

Resources

The book *Neural Networks and Deep Learning* by Michael Nielsen:

`http://neuralnetworksanddeeplearning.com/`

The book *Probabilistic machine learning: An Introduction* by Kevin Murphy:

`https://probml.github.io/pml-book/book1.html`

Colah's blog by Christopher Olah:

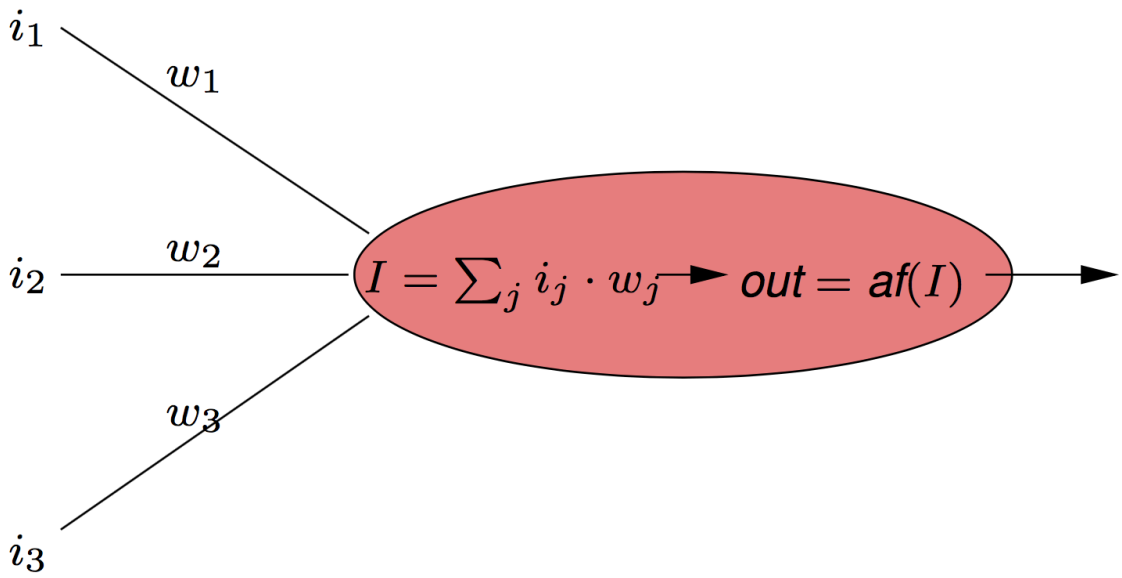
`http://colah.github.io/`

NB:

- Some slides partly based on Ian Goodfellow's lecture slides for the Deep Learning book
- Some figures prepared by Manfred Jaeger

Deep feed forward networks

Neural networks: basic building unit



Two step computation:

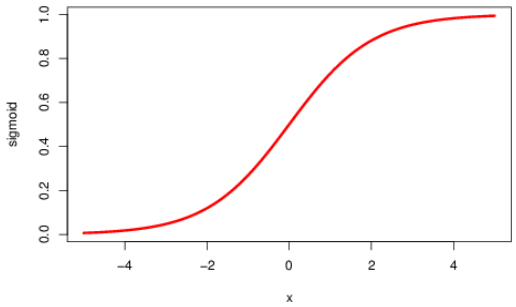
- Combine inputs as *weighted sum* (vector representation: $\mathbf{w}^\mathsf{T}\mathbf{i} = \sum_j i_j w_j$)
- Compute output by **activation function** of combined inputs

Common activation functions

Some common activation functions are:

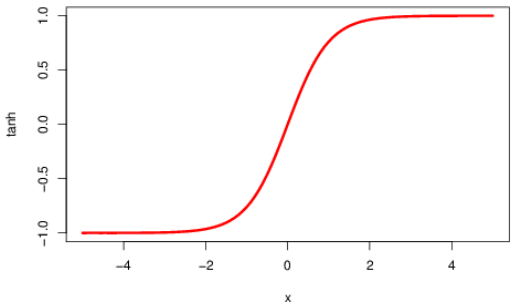
Sigmoid

$af(x) = 1/(1 + e^{-x})$



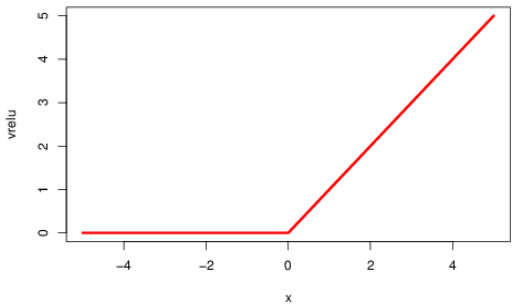
Hyperbolic tangent

$af(x) = \tanh(x)$



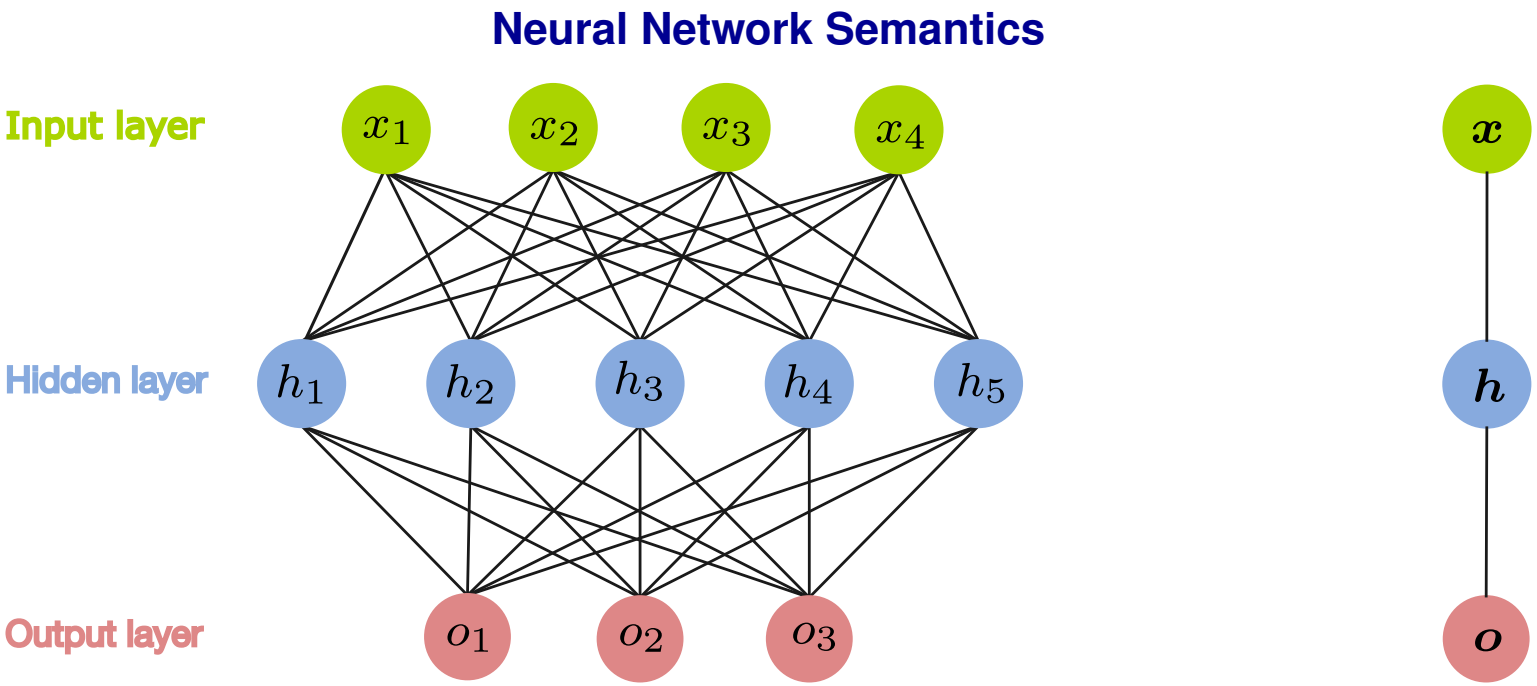
Relu

$af(x) = \max(0, x)$



- If activation function is sigmoid, i.e. $out = \sigma(\sum_j i_j \cdot w_j)$, we also talk of *squashed linear function*.
- For the output neuron also the **identity function** is used: $af(x) = id(x) = x$

Neural network semantics



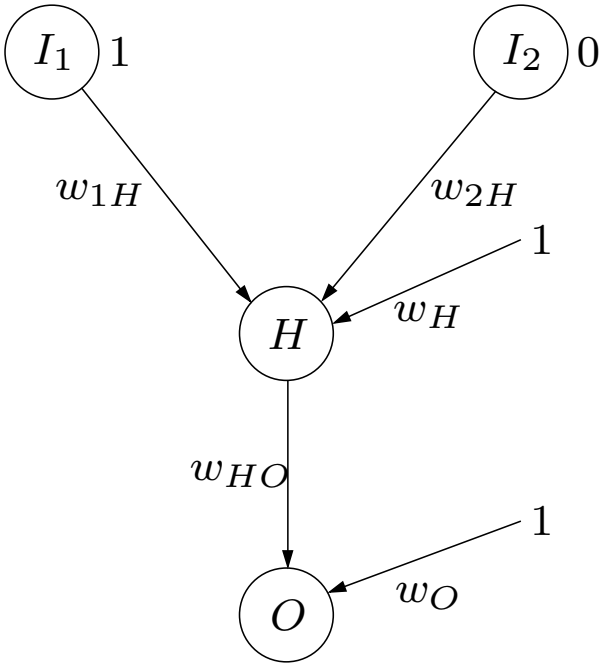
A neural network with n input and k output nodes defines a real-valued function on continuous input attributes:

$$f : \mathbb{R}^4 \rightarrow \mathbb{R}^3.$$

as a composite function:

$$f(\mathbf{x}) = \mathbf{y}$$
$$\mathbf{y} = f_2(W_2^T \mathbf{h} + \mathbf{b}_2)$$
$$\mathbf{h} = f_1(W_1^T \mathbf{x} + \mathbf{b}_1)$$

Propagation in Neural Networks



The input nodes are set to 1 and 0, respectively.

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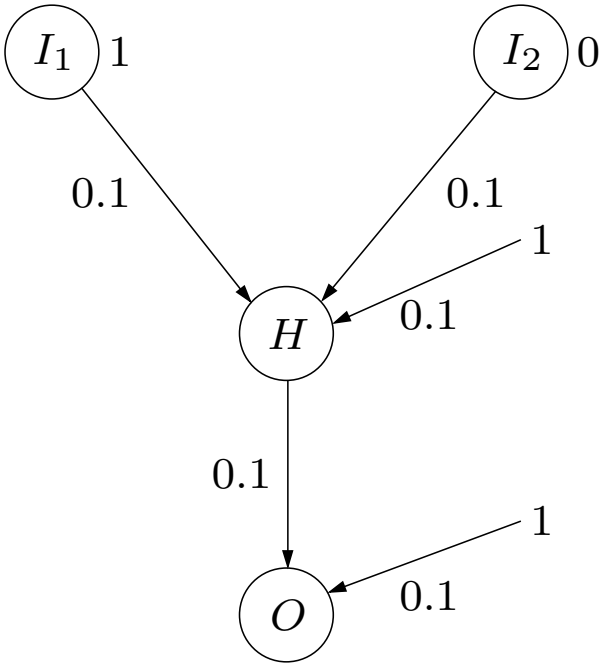

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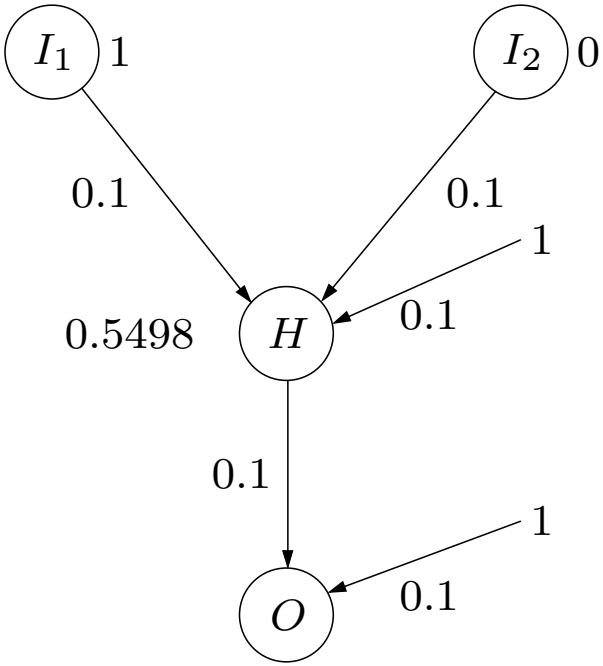
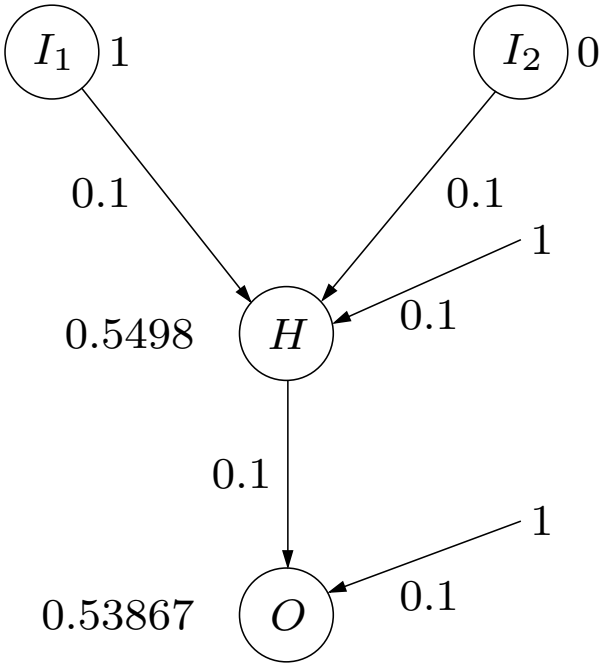


The output of of neuron H is:

$$o_H = \sigma(1 \cdot 0.1 + 0 \cdot 0.1 + 1 \cdot 0.1) = 0.5498.$$

The input nodes are set to 1 and 0, respectively.

Propagation in Neural Networks



The output of of neuron O is:

$$o_H = \sigma(0.5498 \cdot 0.1 + 1 \cdot 0.1) = 0.53867.$$

The input nodes are set to 1 and 0, respectively.

Learning: basics

The Task of Learning

Given:

- the structure and activation functions of the neural network
- the training examples

Input				Output			
X_1	X_2	$\ldots$	X_n	T_1	T_2	$\ldots$	T_m
$x_{1,1}$	$x_{2,1}$	$\ldots$	$x_{n,1}$	$t_{1,1}$	$t_{2,1}$	$\ldots$	$t_{m,1}$
$x_{1,2}$	$x_{2,2}$	$\ldots$	$x_{n,2}$	$t_{1,2}$	$t_{2,2}$	$\ldots$	$t_{m,2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_{1,N}$	$x_{2,N}$	$\ldots$	$x_{n,N}$	$t_{1,N}$	$t_{2,N}$	$\ldots$	$t_{m,N}$

- a *loss* function measuring the error between the target values $\mathbf{t}$ and the output values $\mathbf{o}$:

$$\mathit{loss}(\mathbf{t}, \mathbf{o})$$

Goal: Find the weights $\mathbf{w}$ that minimize the loss over the training examples:

$$\mathit{Loss} = \frac{1}{N} \sum_{i=1}^{N} \mathit{loss}(\mathbf{t}_i, \mathbf{o}_i(\mathbf{w}))$$

Error functions

Squared error

$$\mathit{loss}(\mathbf{t}, \mathbf{o}) = \sum_{i=1}^{m} (t_i - o_i)^2$$

- With a single output neuron, $m = 1$.
- Often used for regression problems.

Log-loss

$$\mathit{loss}(\mathbf{t}, \mathbf{o}) = -\log p(\mathbf{t} \mid \mathbf{x}),$$

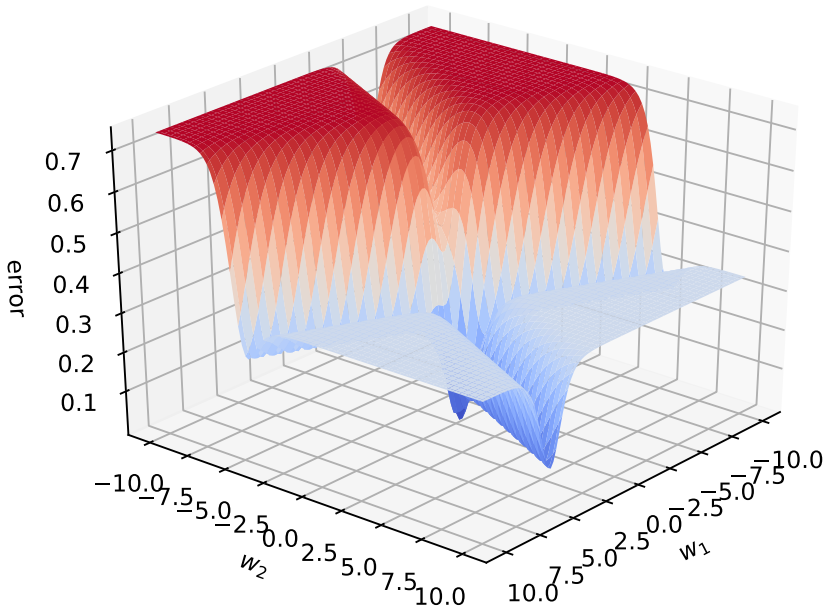
- Specific form of loss depends on p.
- For $p(\mathbf{t} \mid \mathbf{x}) \sim \mathcal{N}(f(\mathbf{x} \mid \mathbf{w}), \mathbf{I})$, loss is similar to squared error.

Cross-entropy

$$\mathit{loss}(\mathbf{t}, \mathbf{o}) = -\sum_{i=1}^{m} t_i \cdot \log(o_i)$$

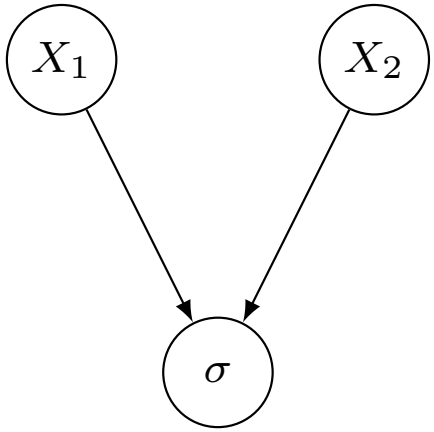
- Outputs should be in the interval $[0, 1]$ (interpreted as probabilities of the m classes).

Error function: example



Error surface with a squared error function

Network structure

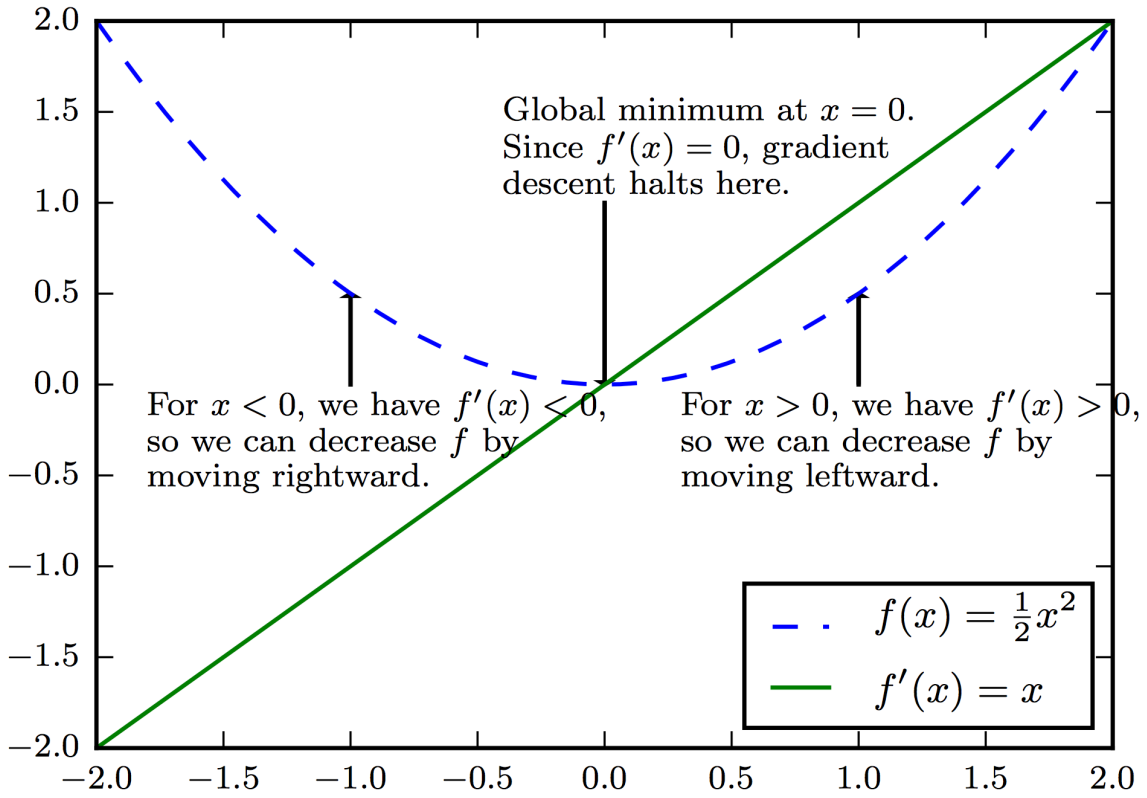


Data

X_1	X_2	O
1	2	0.7
2	−1	0.5

Learning: gradient-based

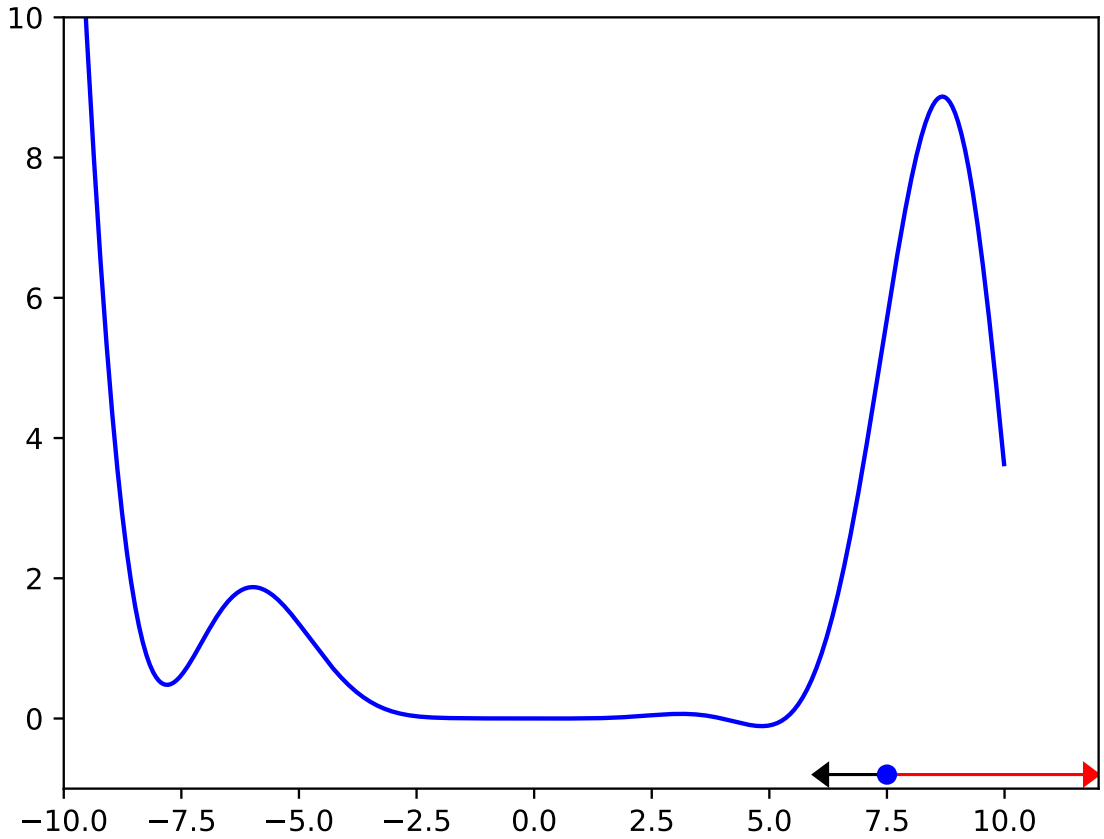
Learning usually involve gradient-based algorithms minimizing a cost function:



Simple gradient descent:

$$\mathbf{w}' = \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathit{Loss}(\mathbf{w})$$

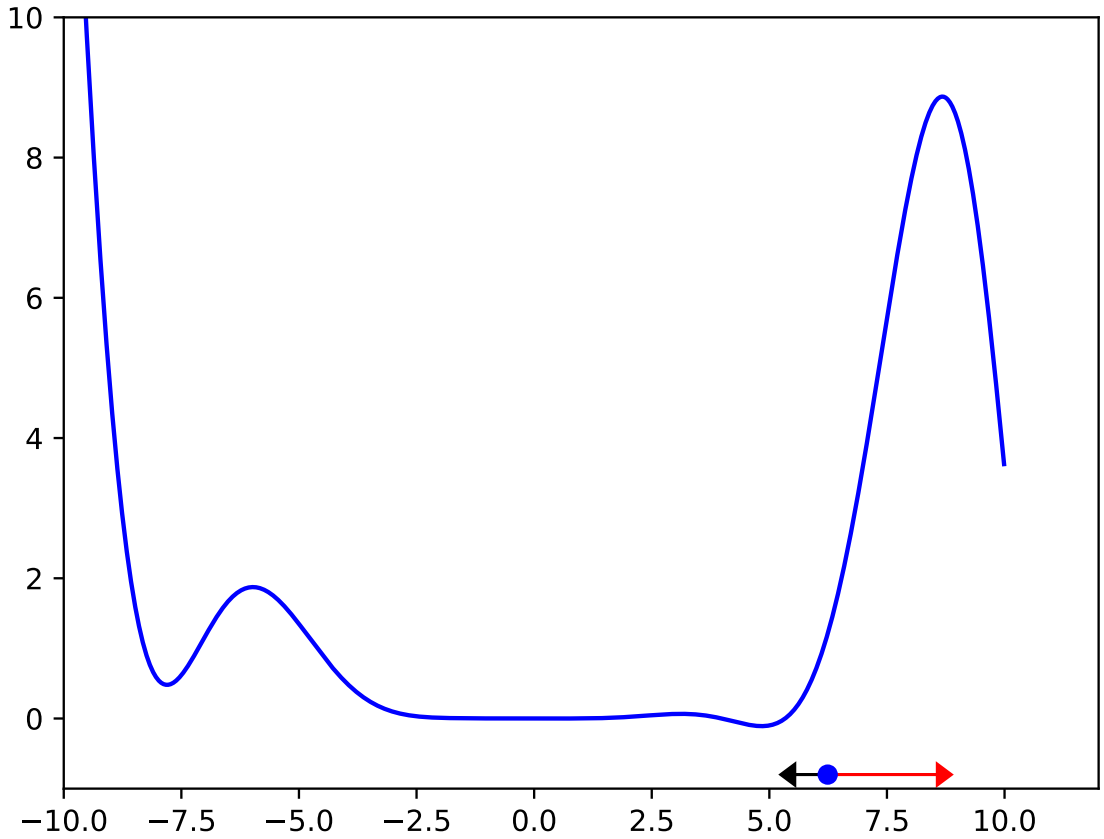
Gradient descent: an example



Gradient descent:

- Pick a random starting point $\mathbf{w}$
- Calculate the gradient/derivative $\nabla \mathit{Loss}(\mathbf{w})$
- Move in the opposite direction of the gradient: $\mathbf{w}' = \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathit{Loss}(\mathbf{w})$
- Until convergence . . . (convergence depends on learning rate η)

Gradient descent: an example



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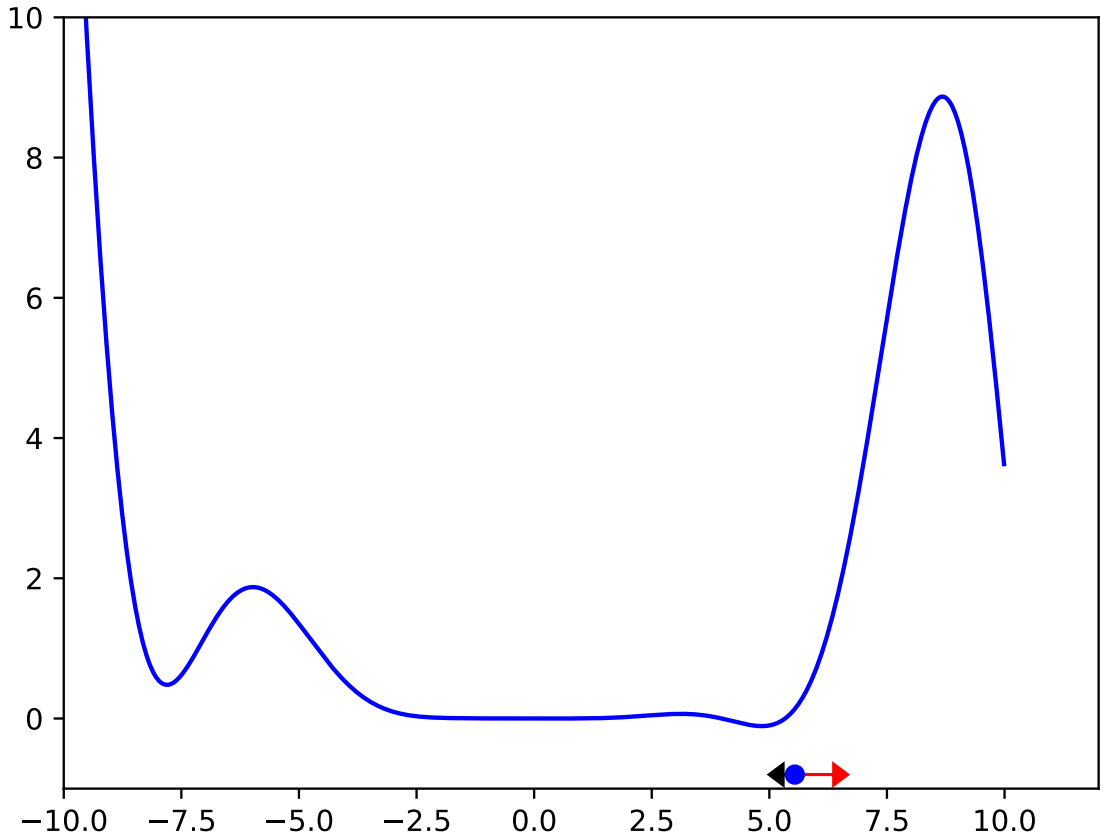

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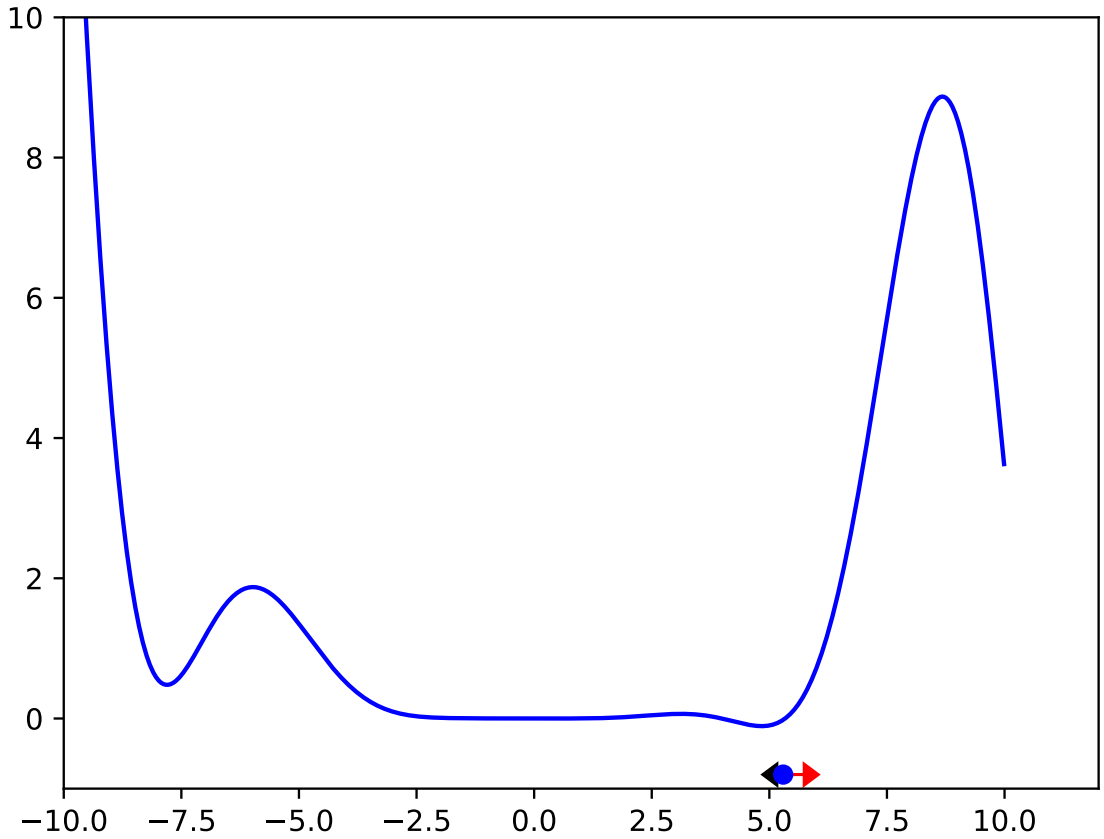

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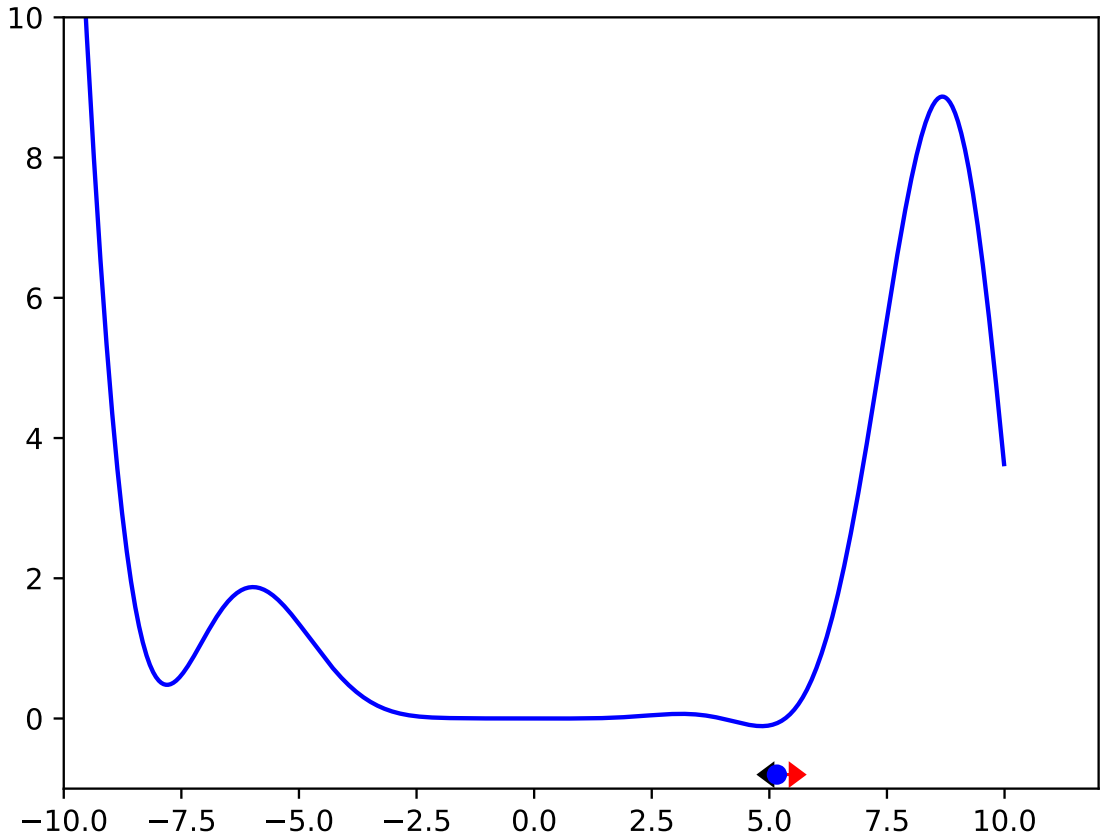

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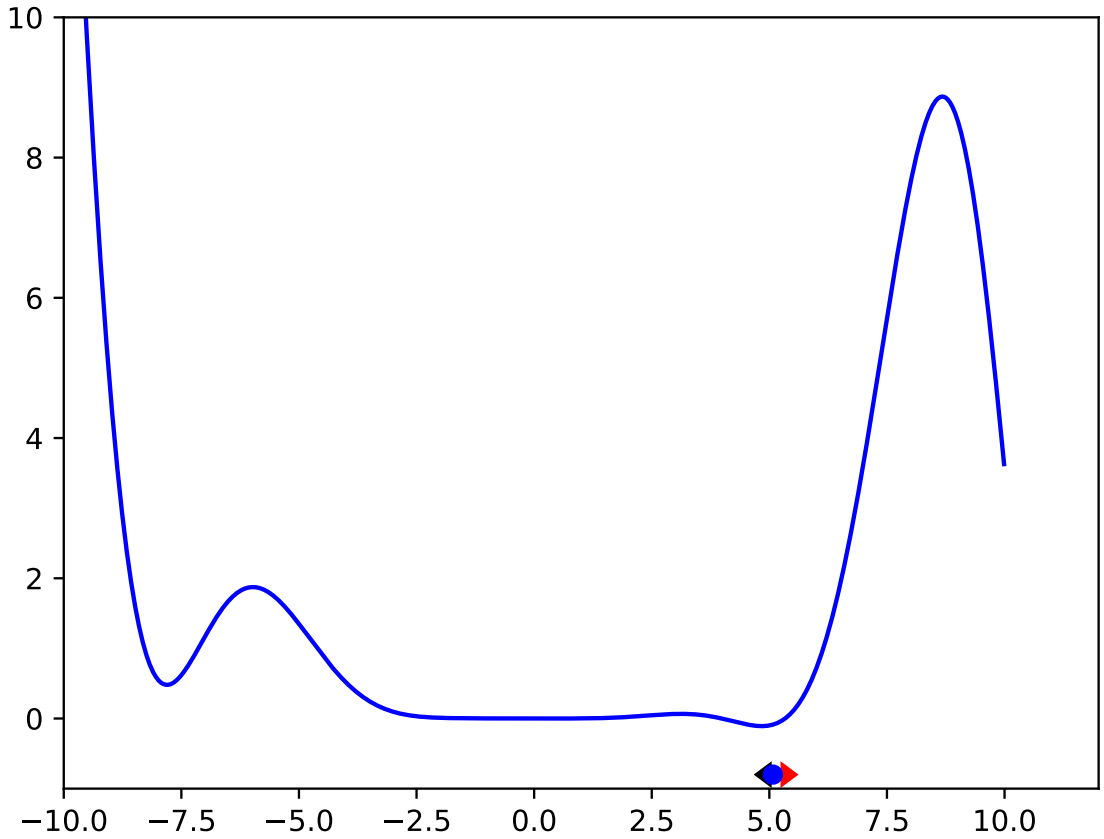

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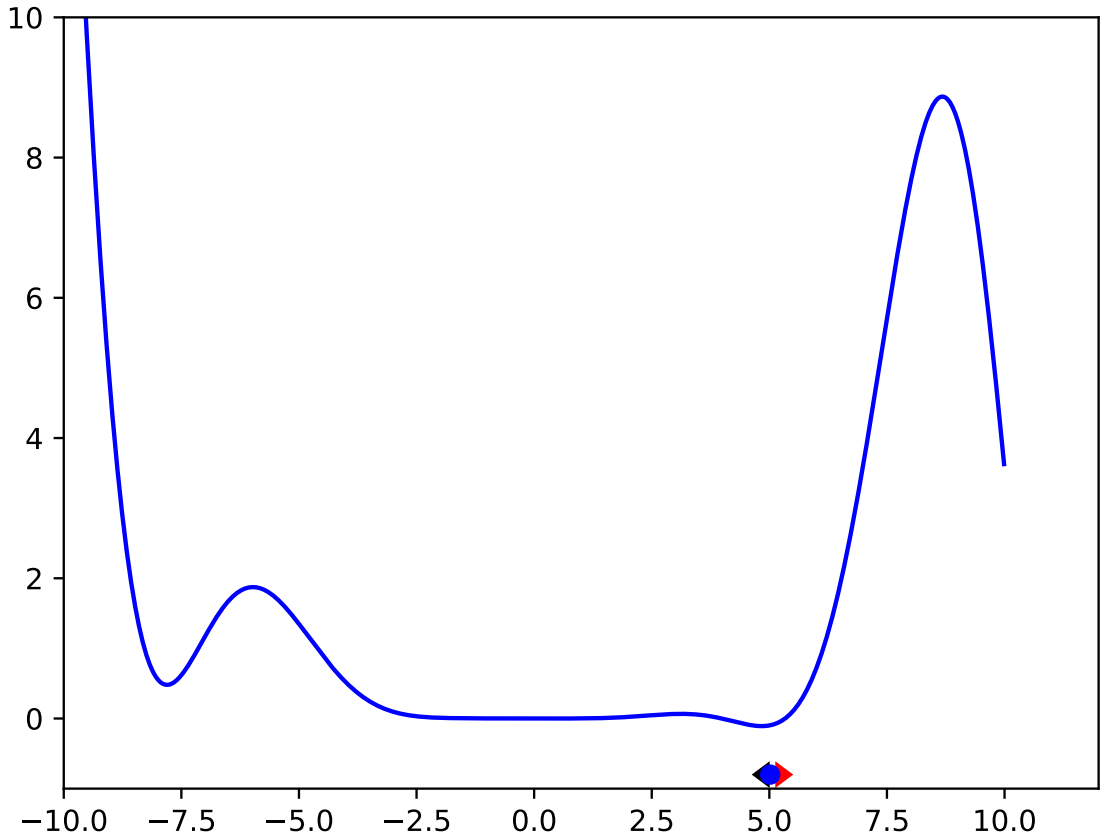

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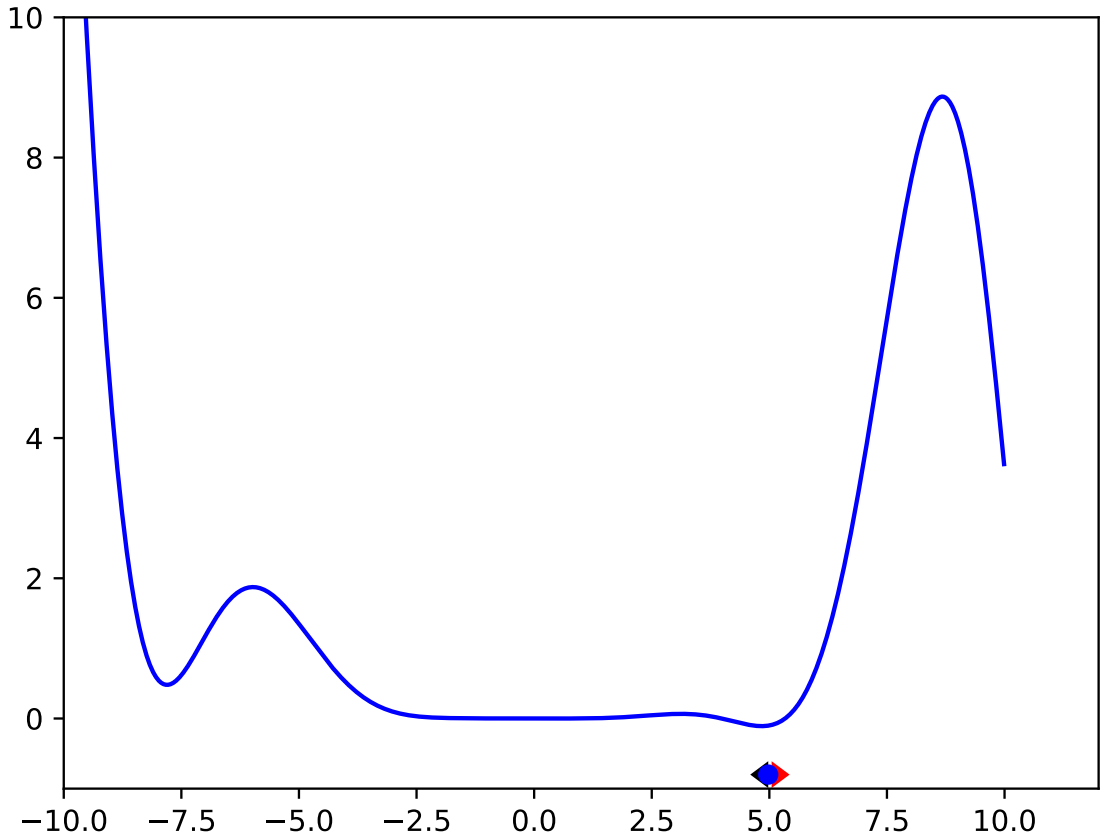

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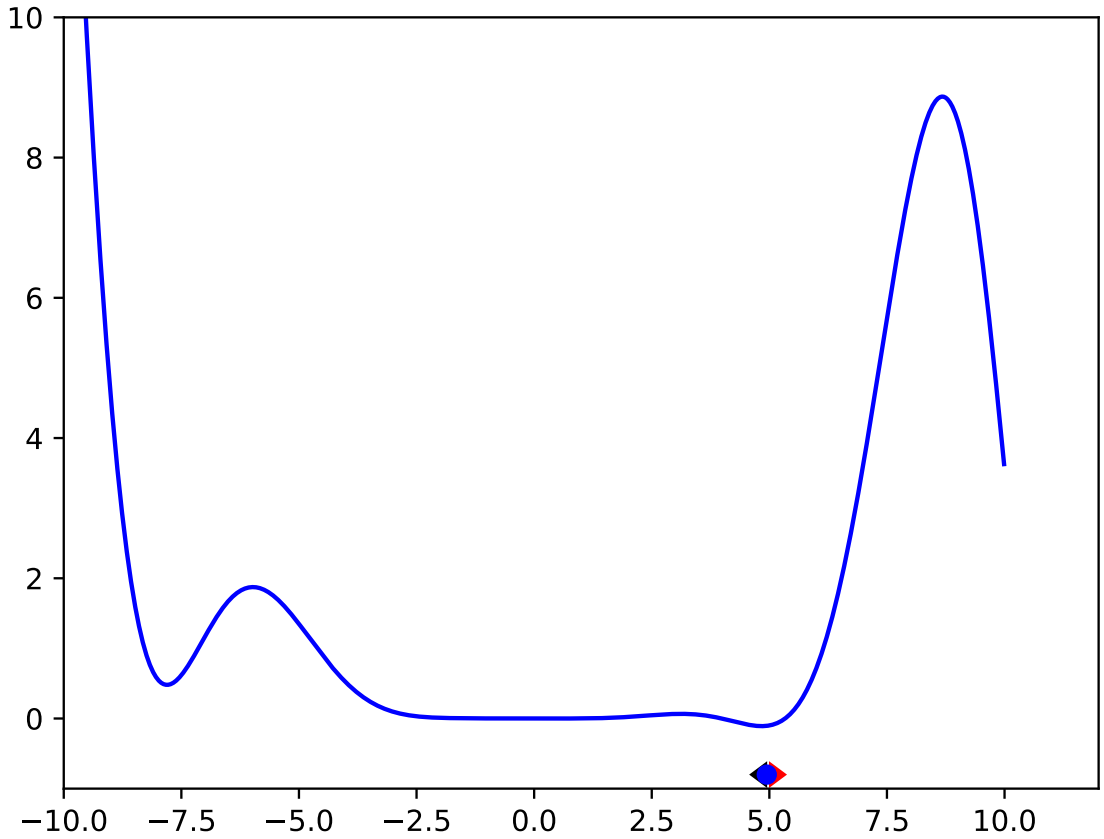

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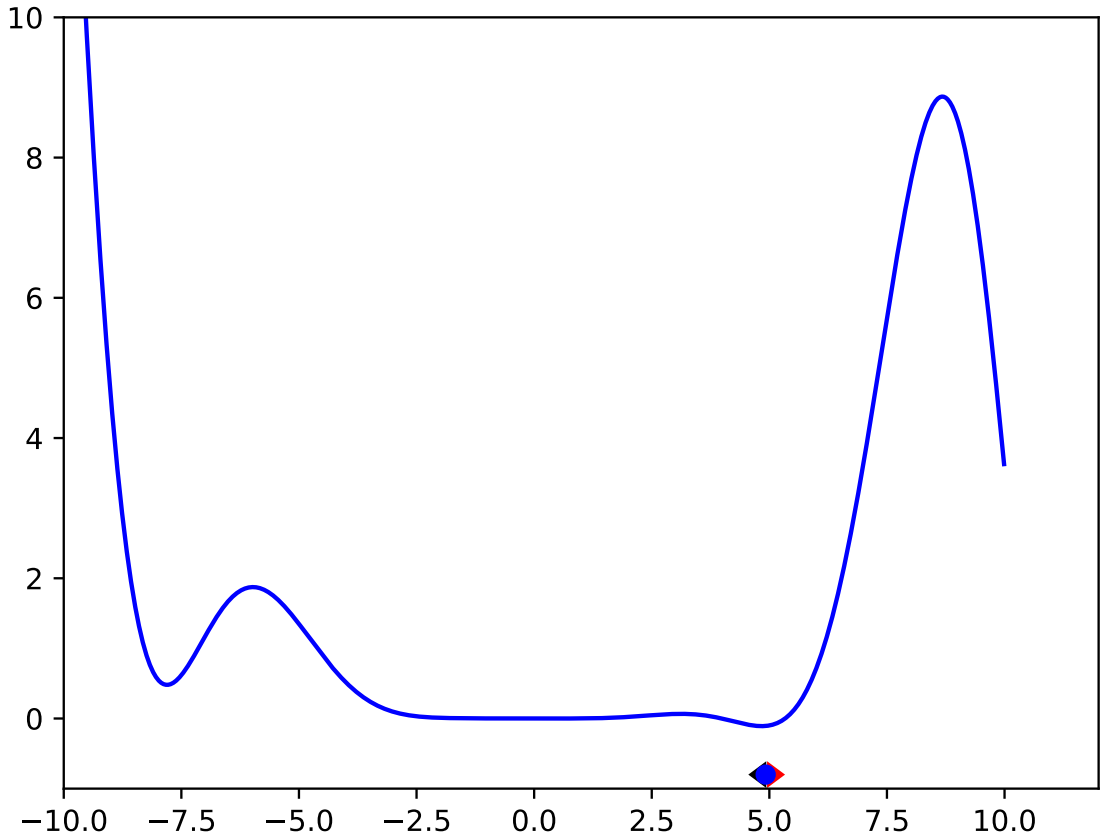

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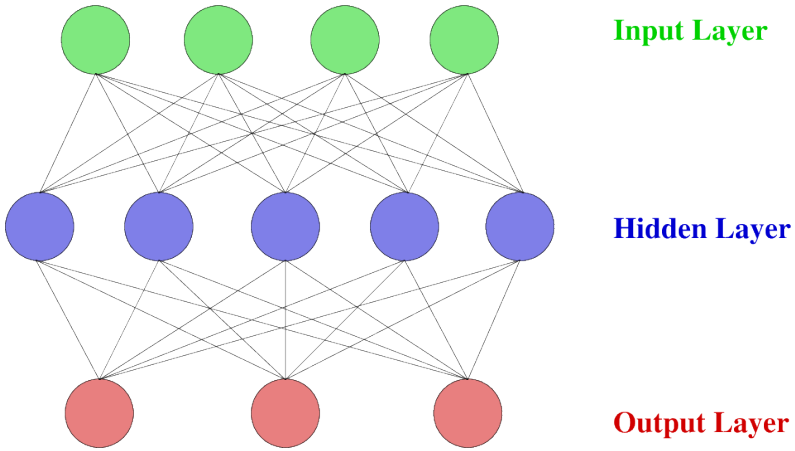
Back propagation

Back propagation

Training examples provide target values for only network outputs, so no target values are directly available for indicating the error of the hidden units' values.



$\rightsquigarrow$ compute local error terms for hidden nodes and adjust weights (through partial derivatives) using local terms.

Back propagation

An algorithm that computes derivaties using the chain rule.

The scalar case

$$\frac{\partial f(g(x))}{\partial x} = \frac{\partial f(g(x))}{\partial g(x)} \frac{\partial g(x)}{\partial x}$$

With multiple component functions

$$\frac{\partial f(g_1(x), g_2(x))}{\partial x} = \frac{\partial f(g_1(x), g_2(x))}{\partial g_1(x)} \frac{\partial g_1(x)}{\partial x} + \frac{\partial f(g_1(x), g_2(x))}{\partial g_2(x)} \frac{\partial g_2(x)}{\partial x}$$

Example

For the function $f(x) = \sigma(x) \cdot x^2$ we have

$$\begin{aligned} \frac{\partial f(x)}{\partial x} &= \frac{\partial f(x)}{\partial \sigma(x)} \frac{\partial \sigma(x)}{\partial x} + \frac{\partial f(x)}{\partial x^2} \frac{\partial x^2}{\partial x} \\ &= x^2 \cdot \sigma'(x) + 2\sigma(x) \cdot x \end{aligned}$$

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A note on notation: Vector valued functions

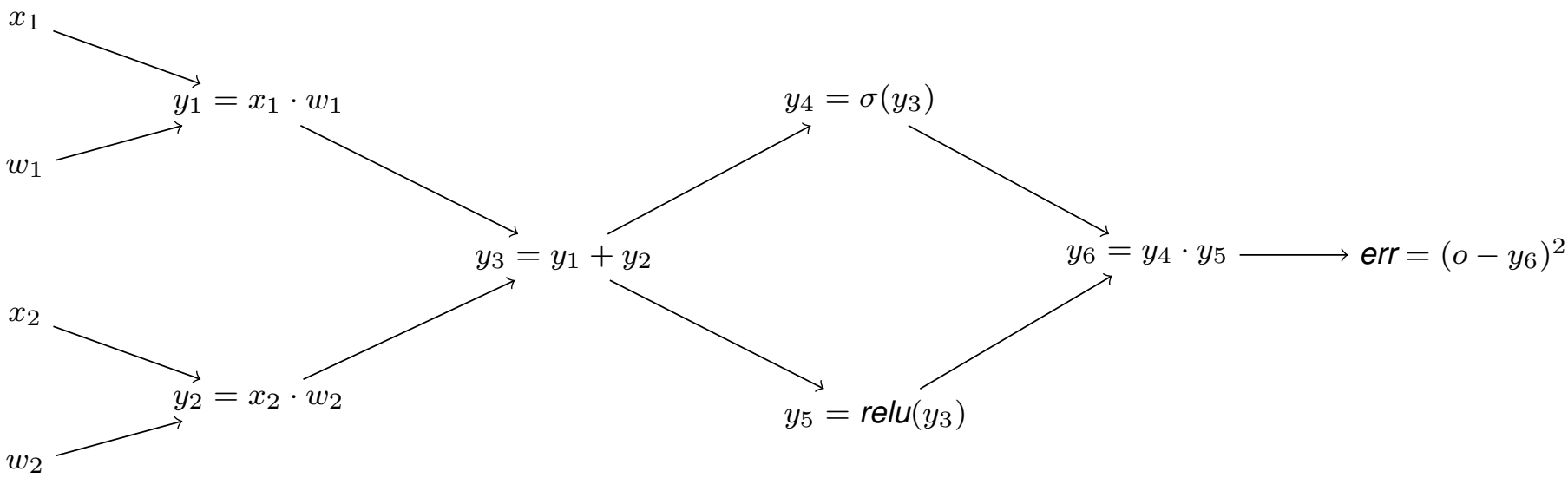
For $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$, if $\mathbf{y} = g(\mathbf{x})$ and $\mathbf{z} = f(g(\mathbf{x})) = f(\mathbf{y})$, then

$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}$$

In vector notation: $\nabla_{\mathbf{x}} z = \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right)^{\mathsf{T}} \nabla_{\mathbf{y}} z$, where $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ is the $n \times m$ Jacobian matrix.

Example

Computational graph



Forward pass

- Initialize and perform a full propagation through the network

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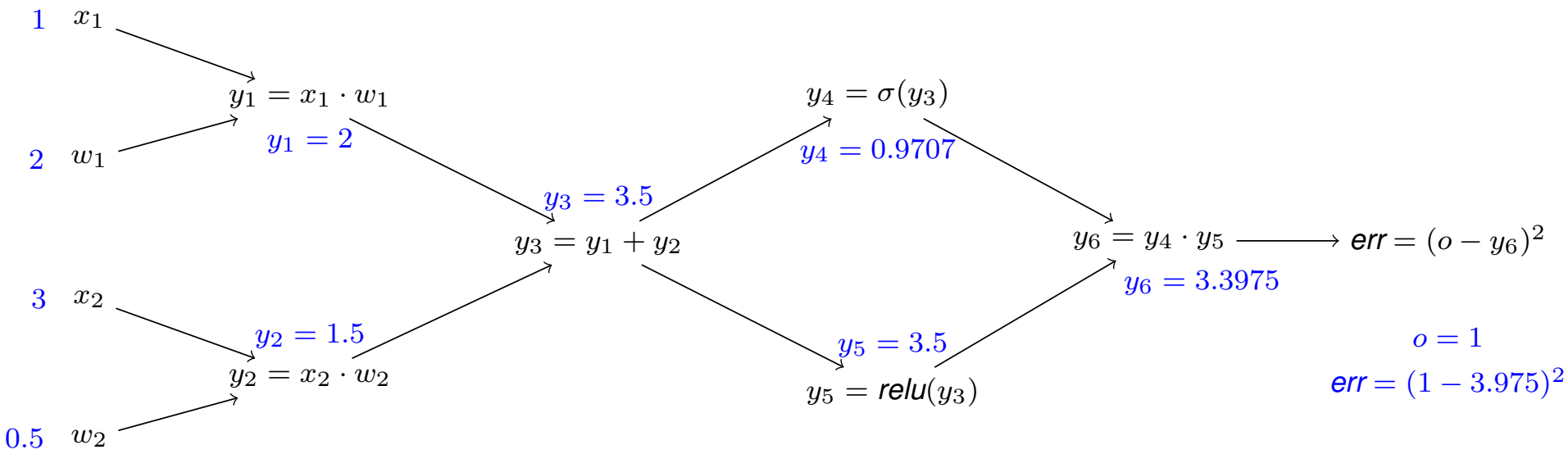

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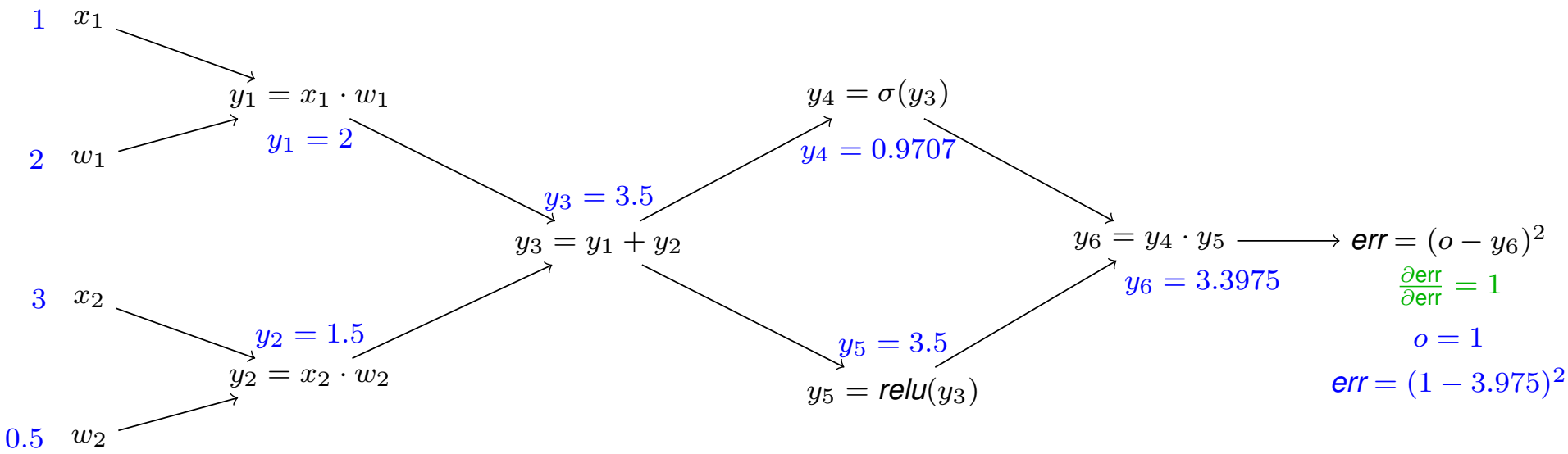

Back propagation Application of

- Chain rule
- Dynamic programming

Recall chain rule

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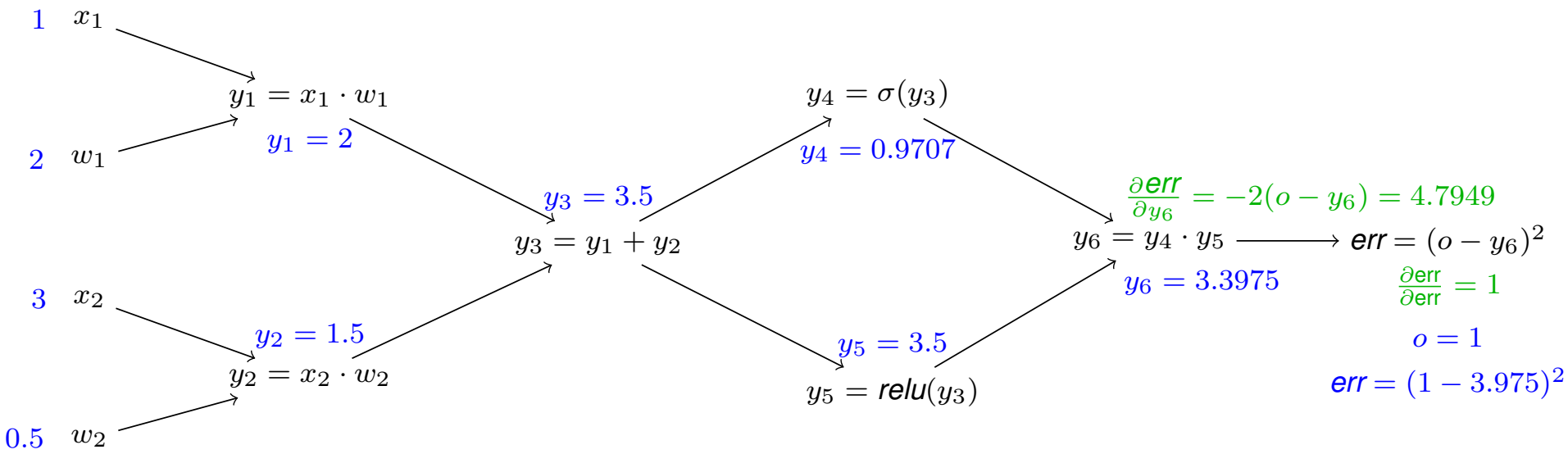

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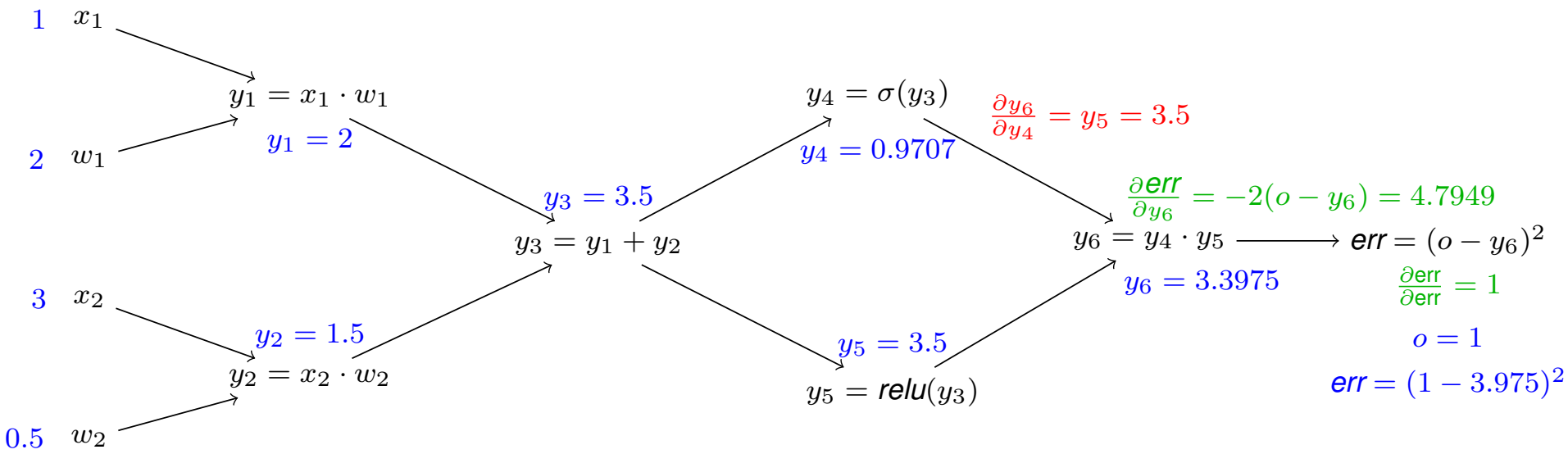

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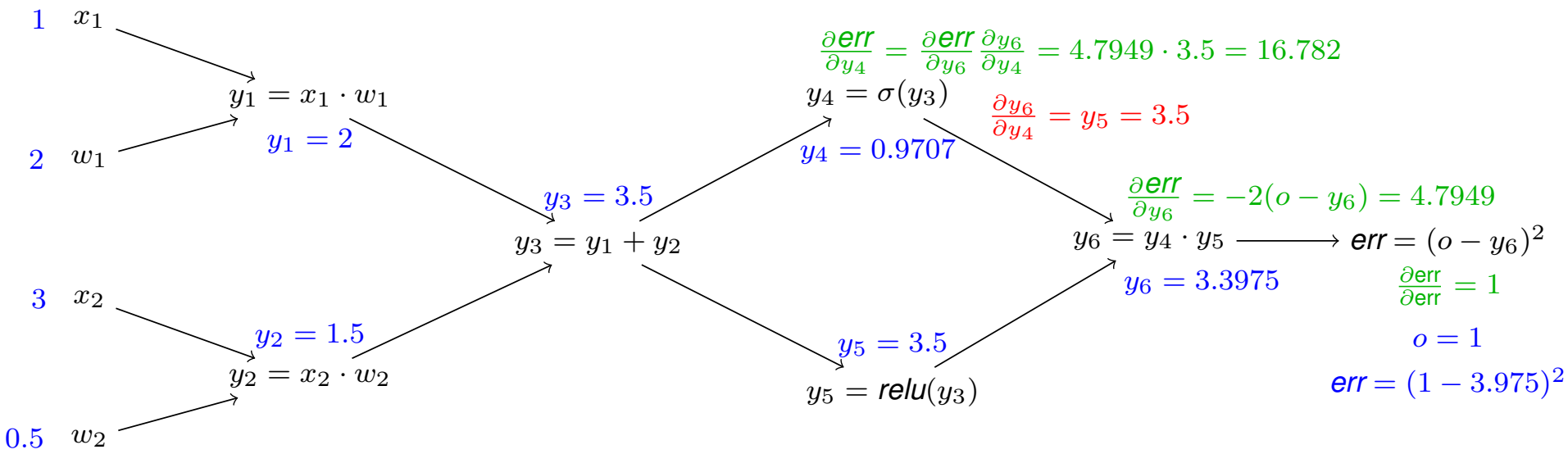

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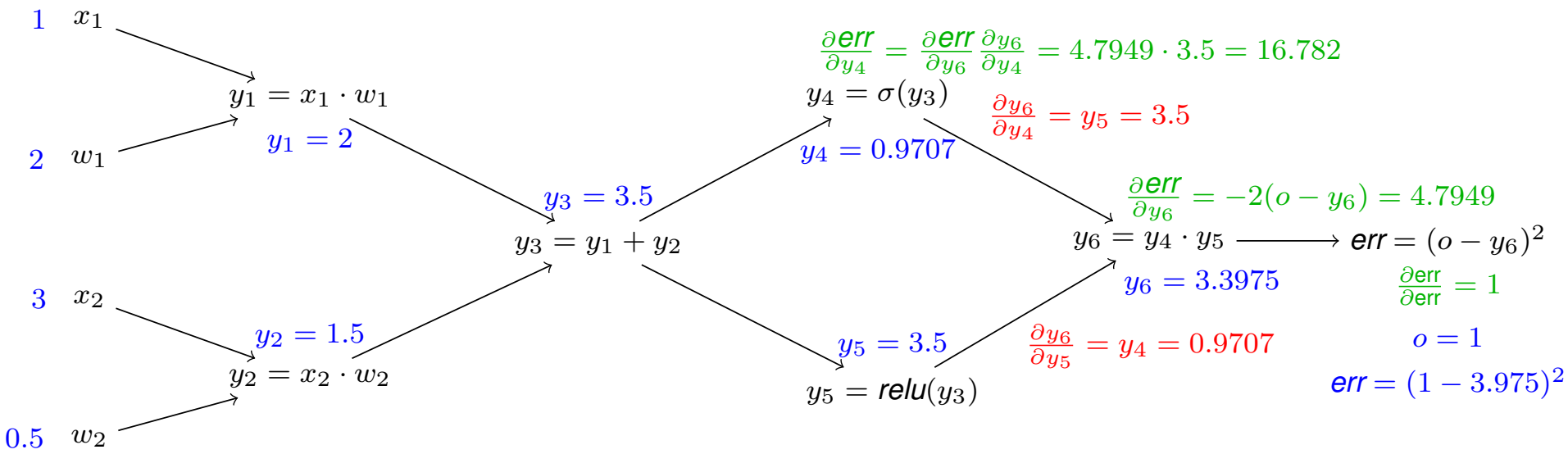

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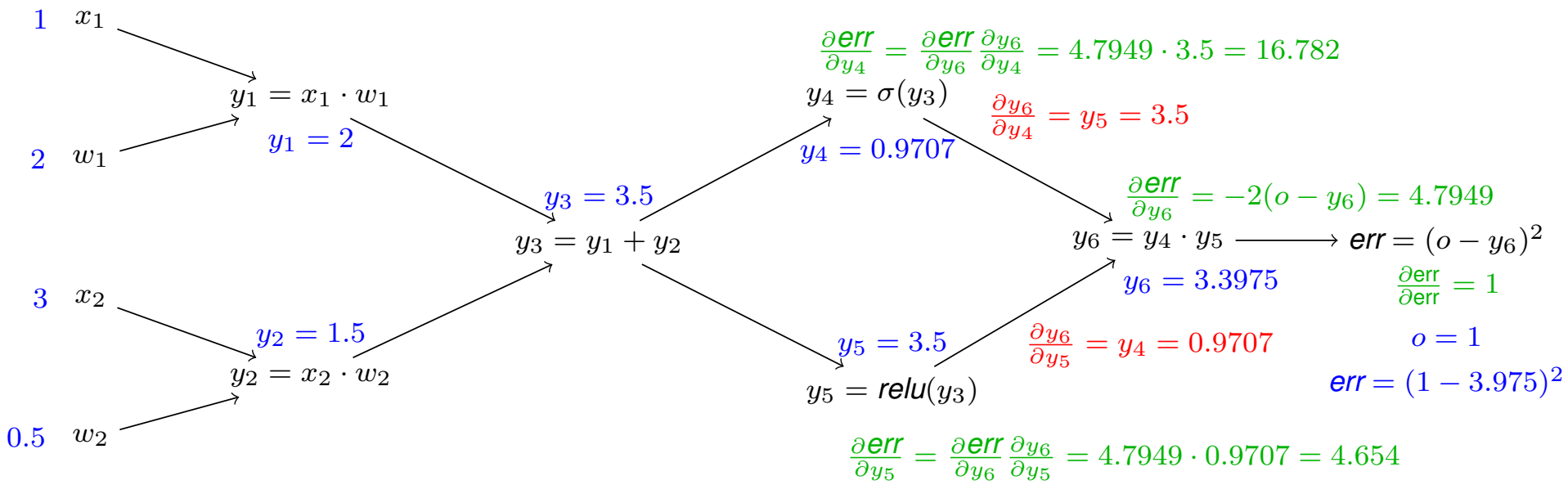

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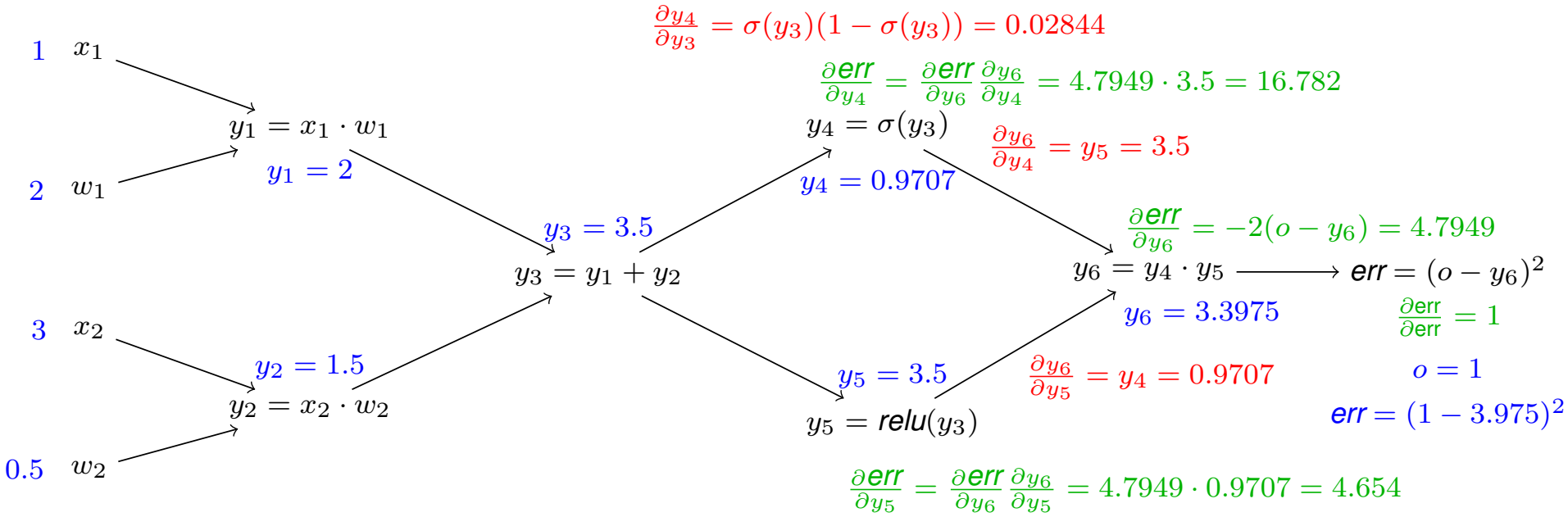
Example

Computational graph



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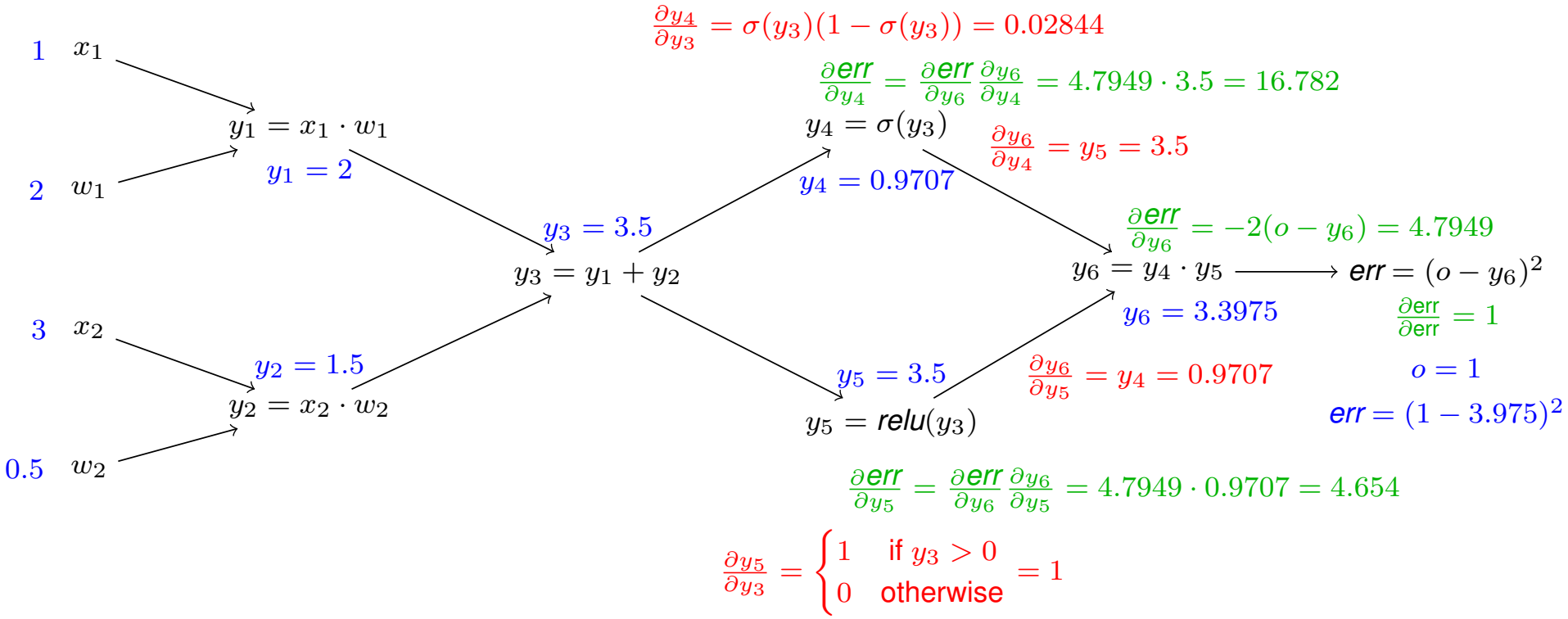

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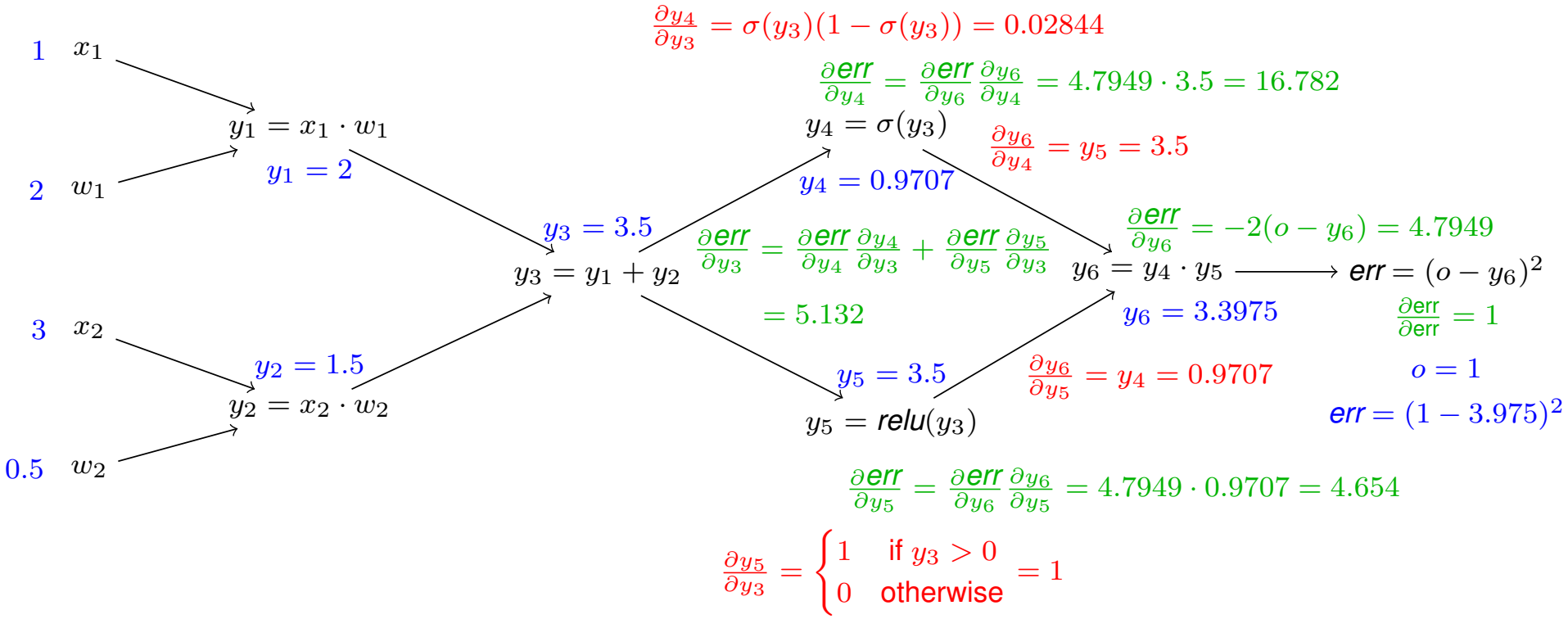

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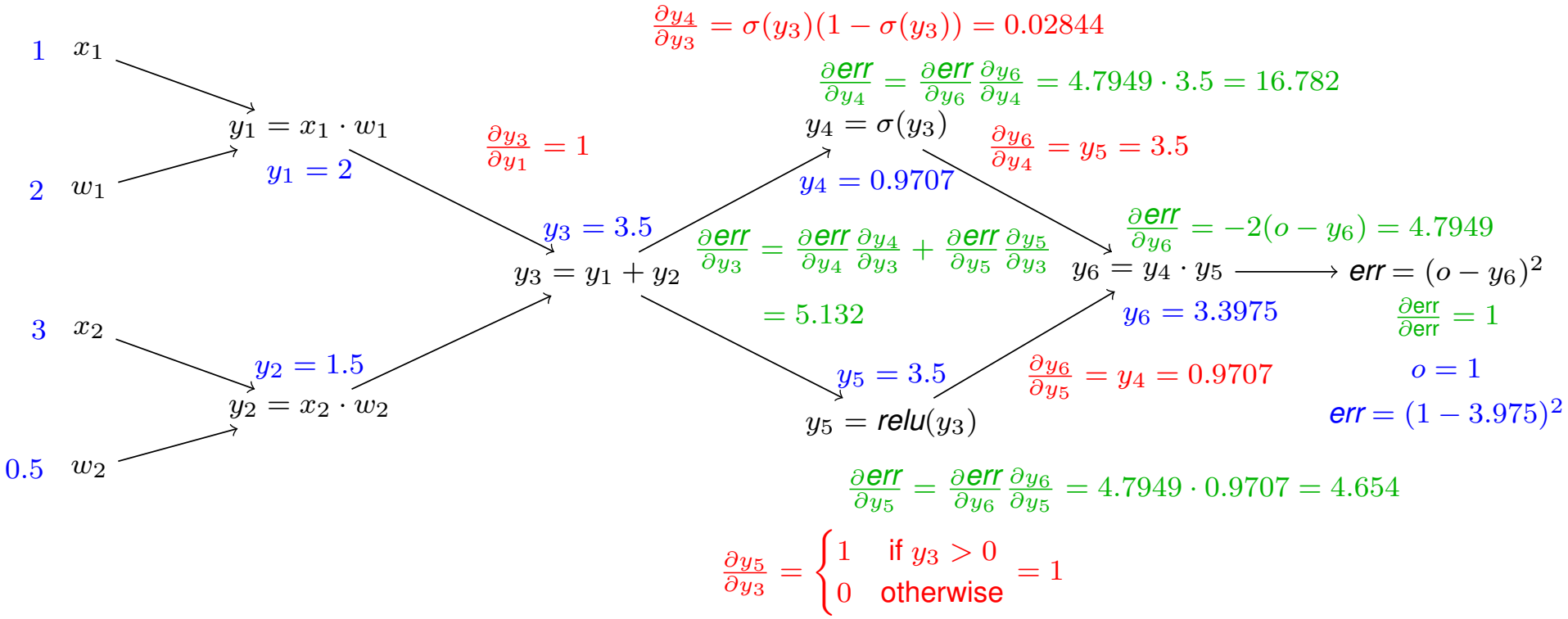

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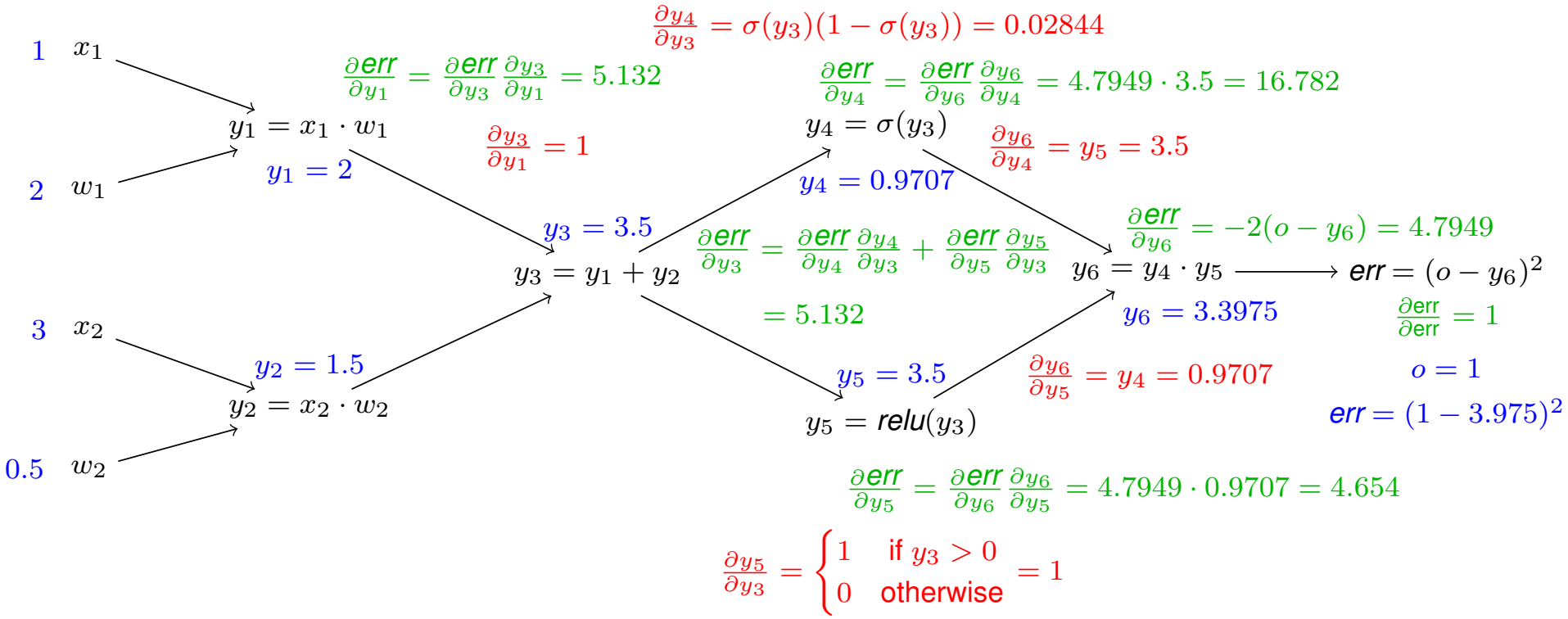

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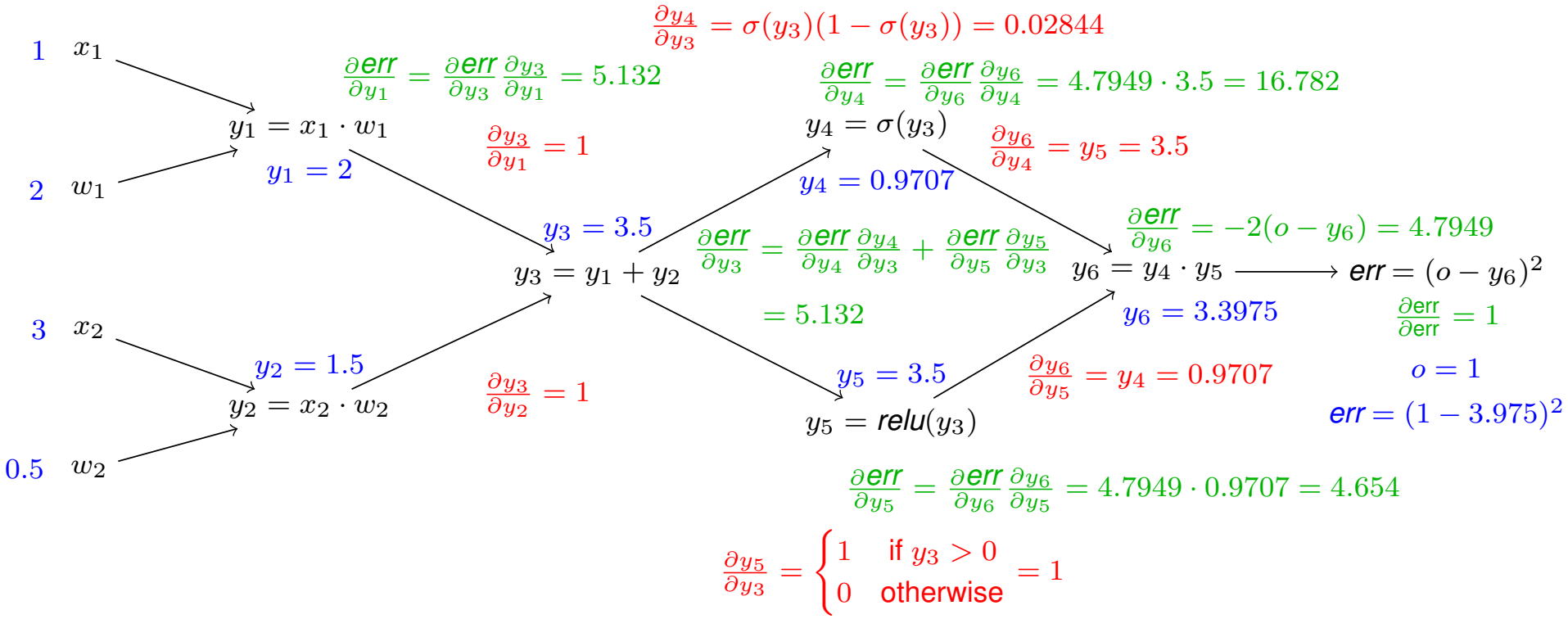

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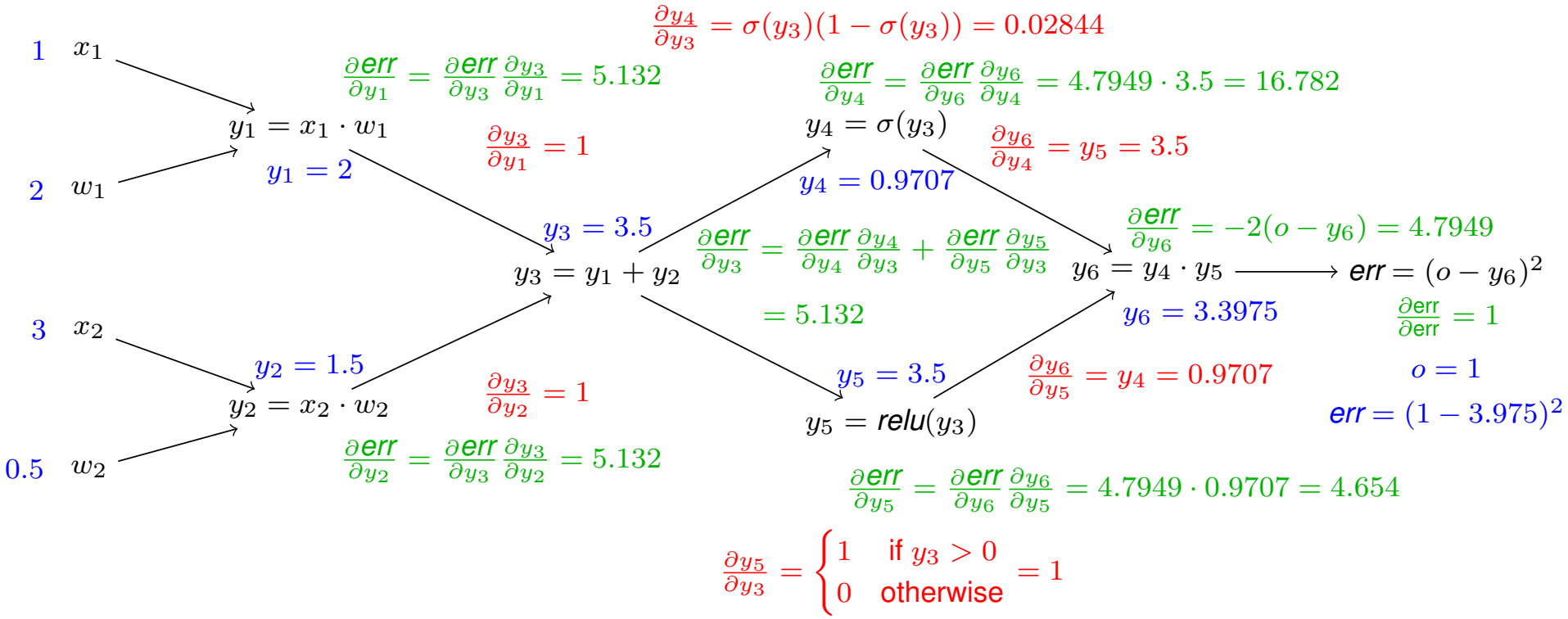

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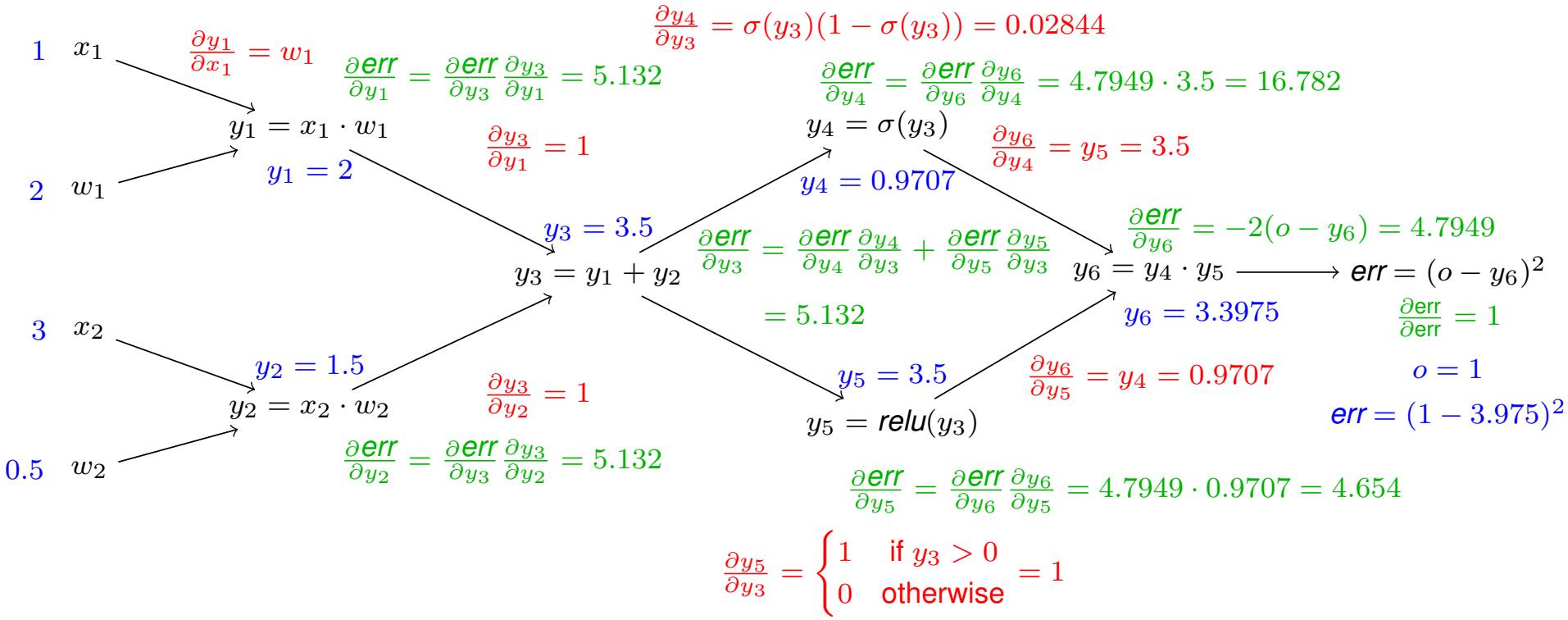

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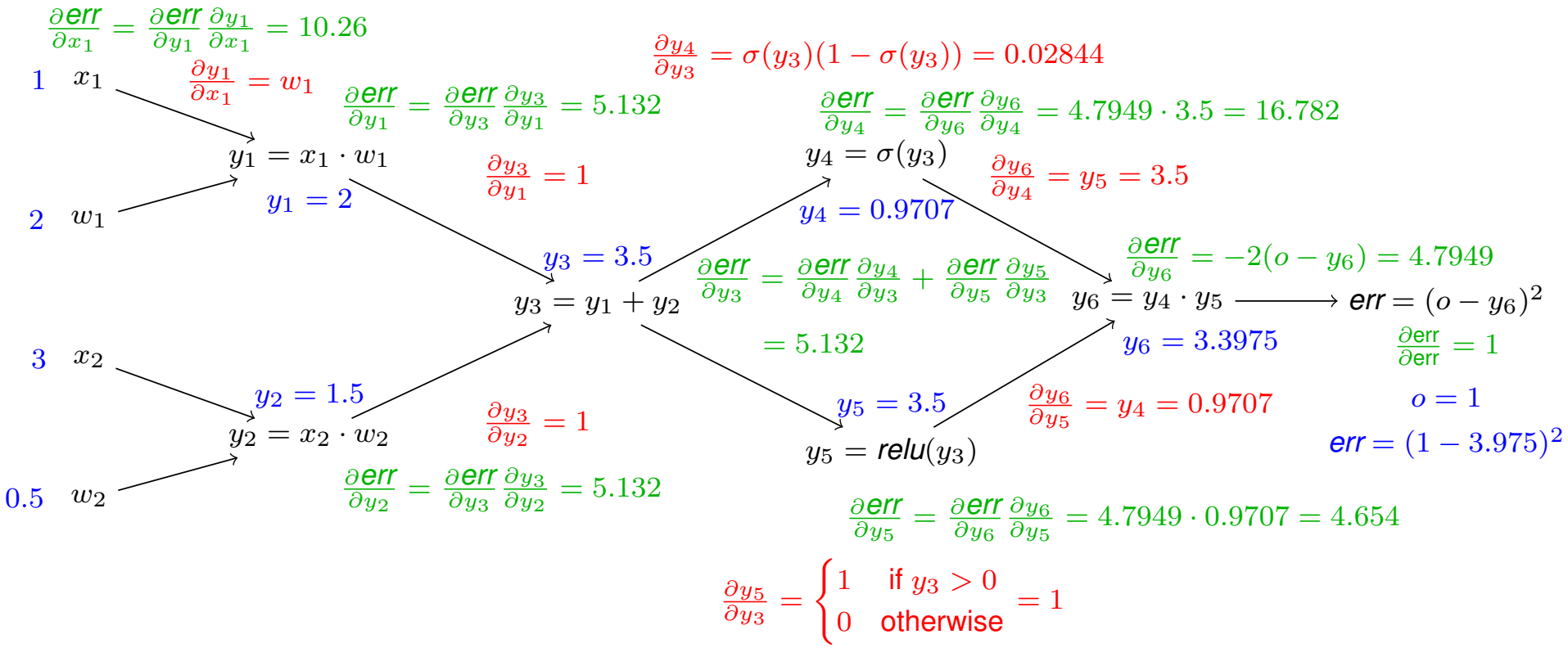

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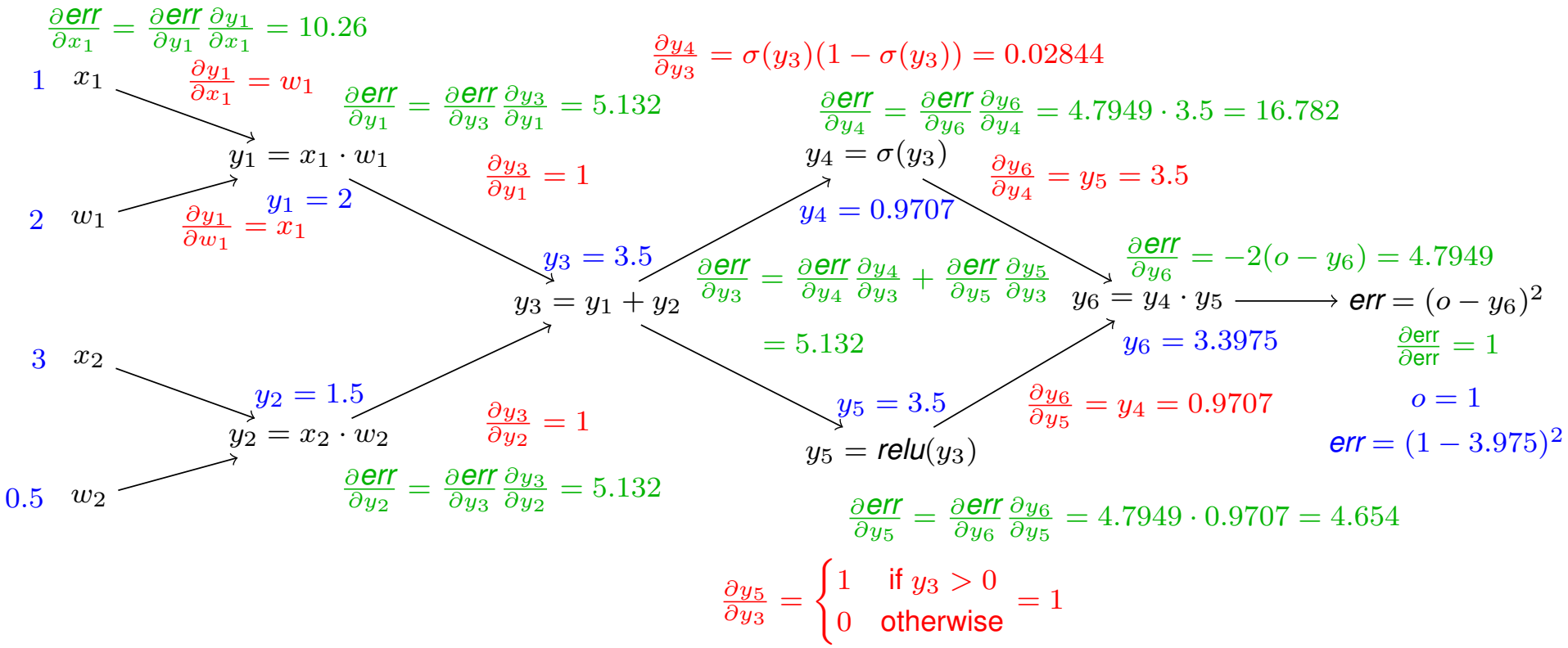

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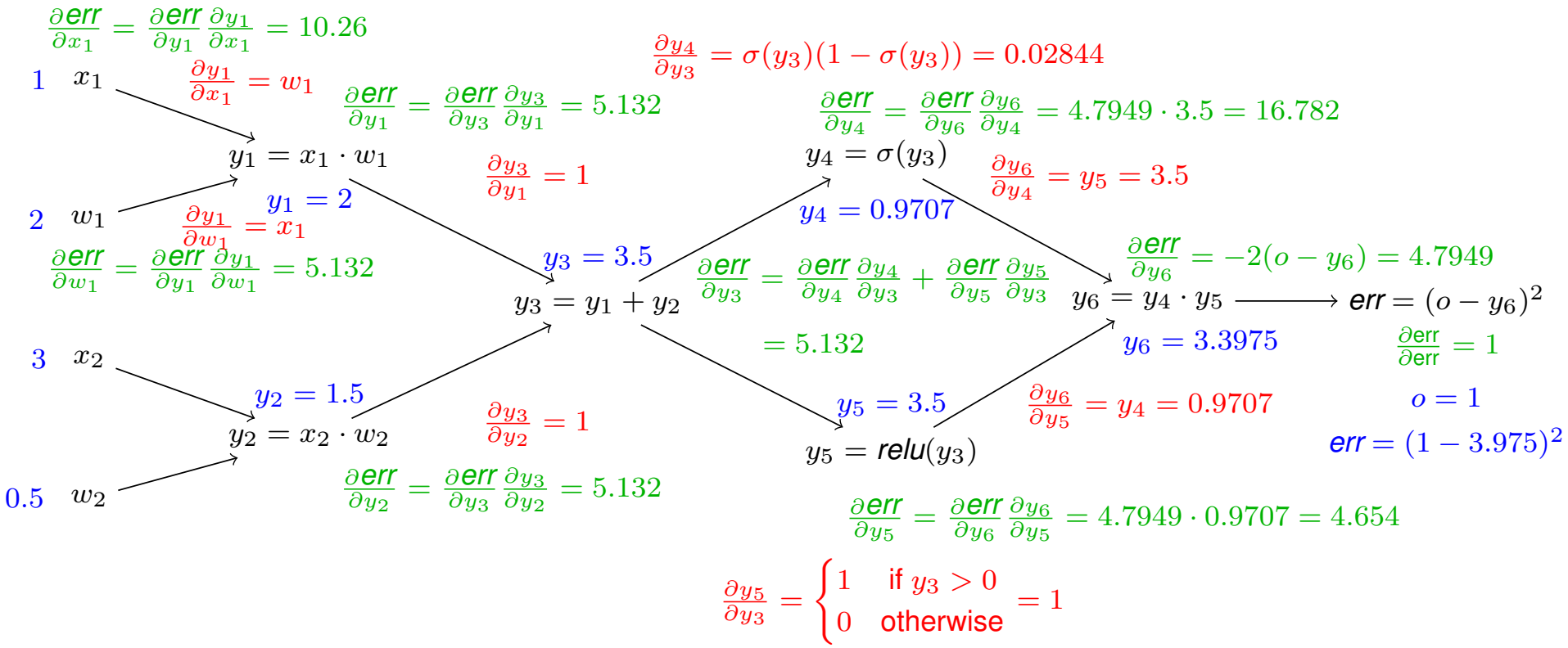

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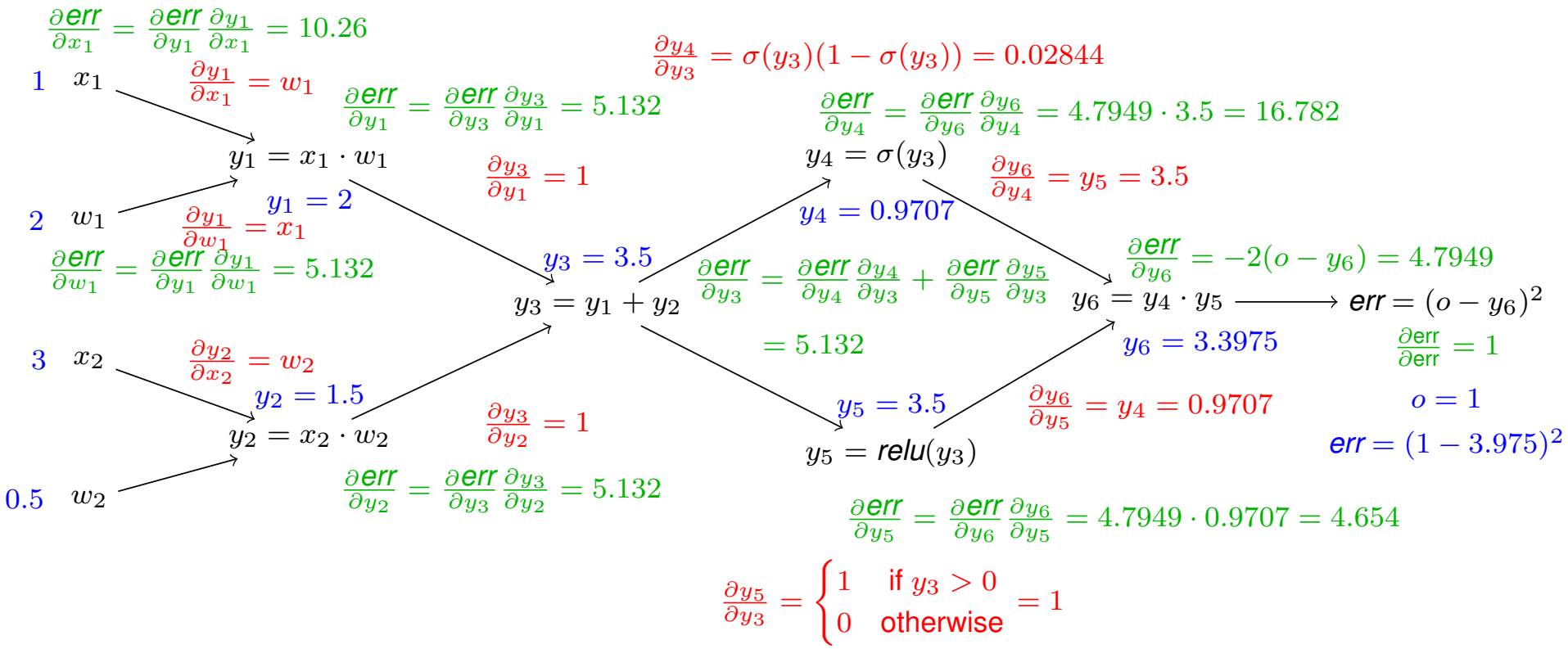

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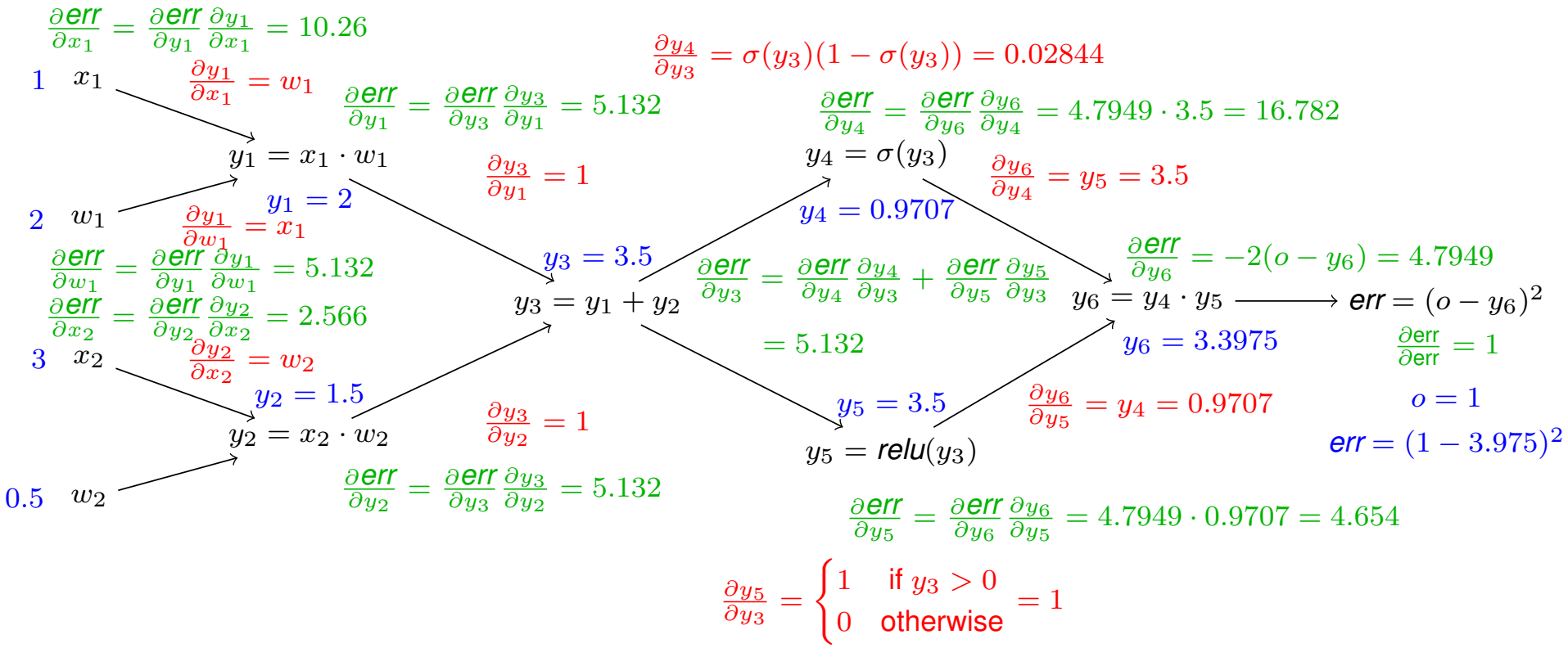

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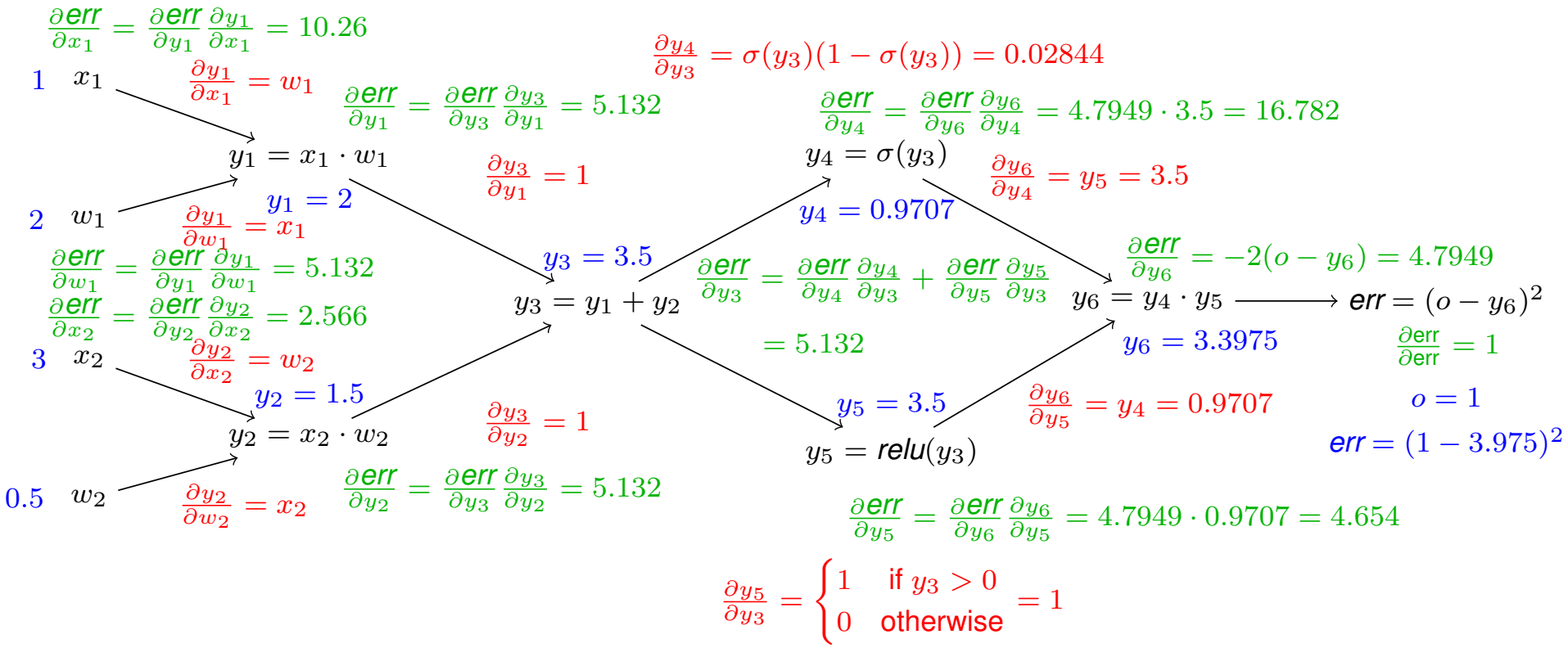

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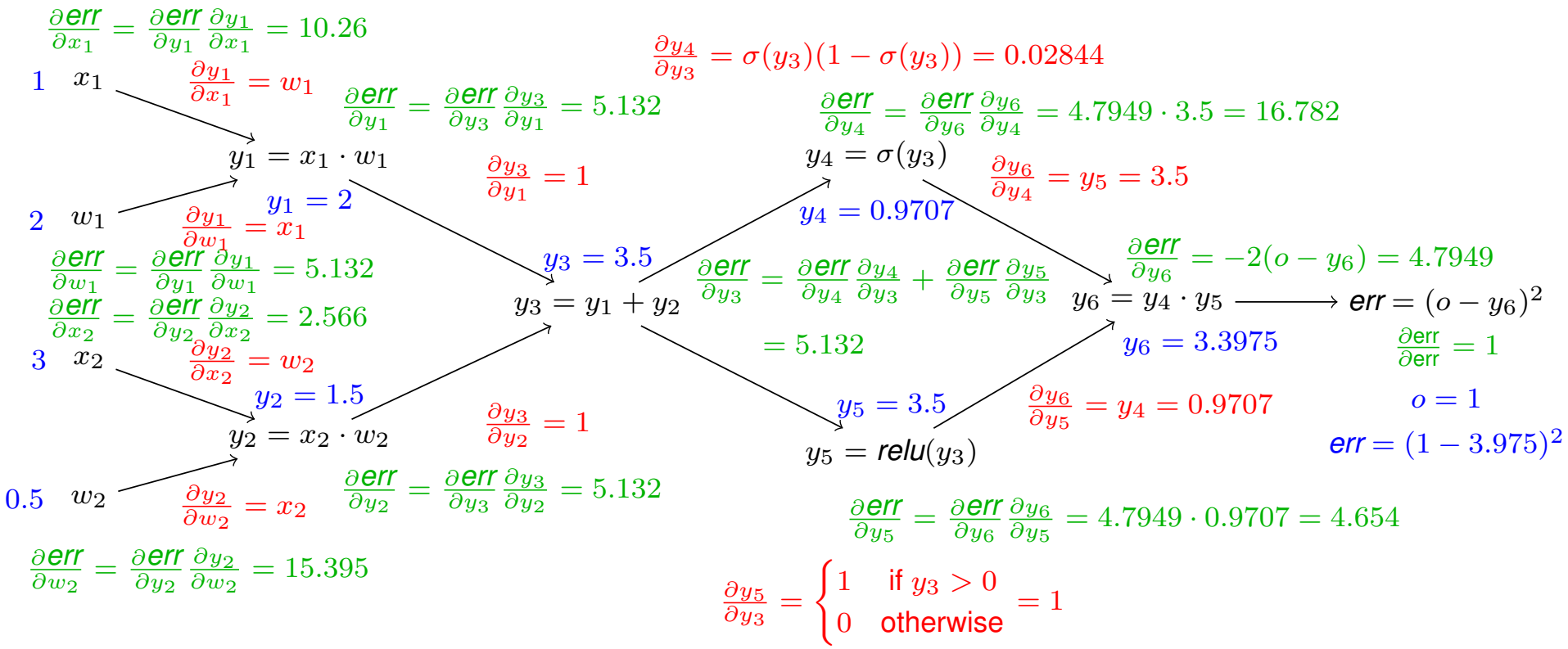

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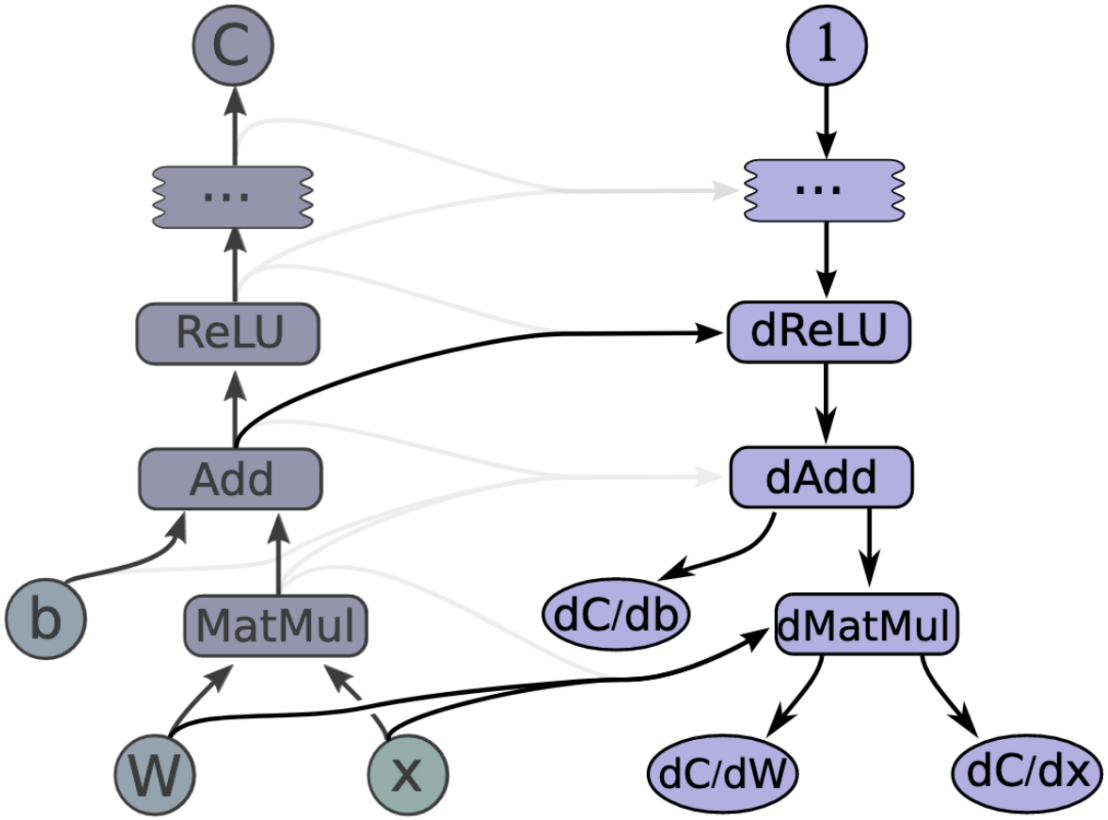
Back propagation and computational graphs

Implementing back propagation only requires the specification of

- the network structure
- the derivatives of the operations performed



Abadi, Martiín, et al. “Tensorflow: Large-scale machine learning on heterogeneous distributed systems.” (2016).

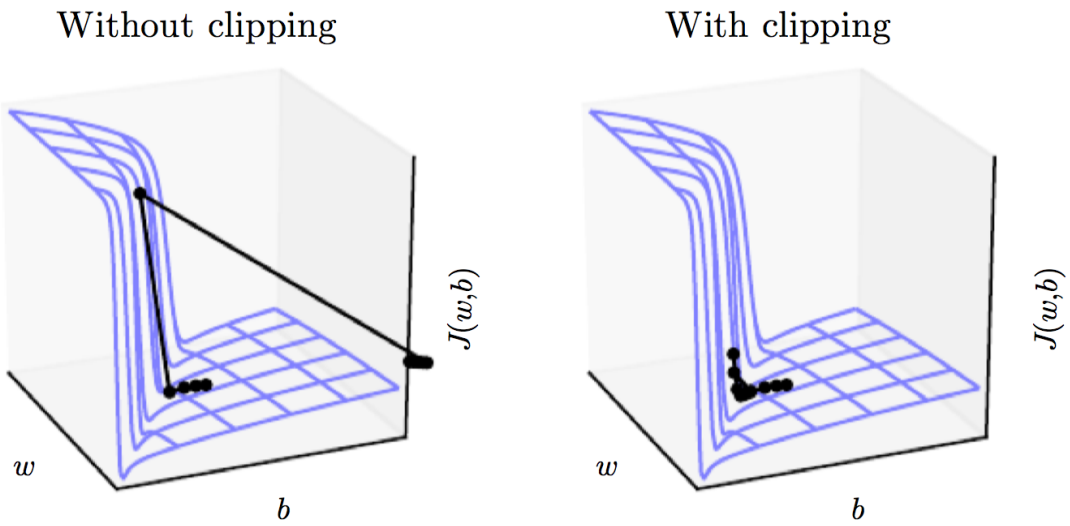
Jupyter notebook

gradient_example.py

More on derivatives and descent-based learning

Cliffs and exploding gradients

Neural networks with many layers can have steep regions in the cost landscape:



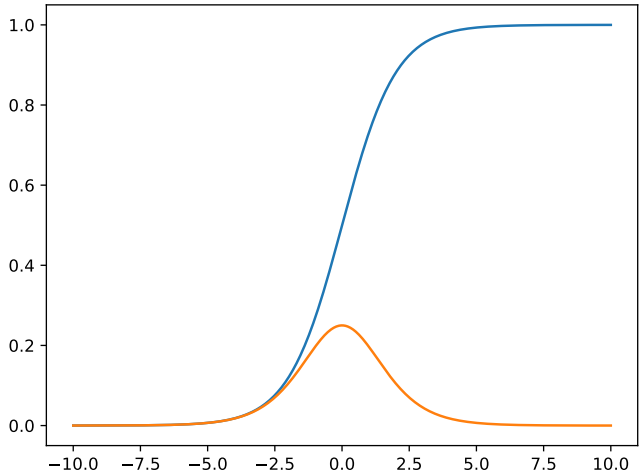
Idea

Clip the gradient before parameter updating:

$$\text{if } \|\mathbf{g}\| > v \text{ then } \mathbf{g} \leftarrow \frac{\mathbf{g}v}{\|\mathbf{g}\|}$$

Vanishing gradients

The sigmoid function and its derivative



If the weights are initialized too large:

- $\sigma(x)$ will be close to either 0 and 1
- $\frac{\partial}{\partial x}\sigma(x) = \sigma(x) \cdot (1 - \sigma(x))$ will be close to 0

$\rightsquigarrow$ gradients will vanish and the rest of the backward pass will be affected due to the multiplication in the chain rule.

Gradient-based learning

The cost function often decomposes over into a sum over the training examples

$$\mathit{Loss}(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N} \mathit{loss}(\mathbf{x}^{(i)}, y^{(i)} \mid \mathbf{w})$$

For the log-loss: $\mathit{loss}(\mathbf{x}^{(i)}, \mathbf{y}^{(i)} \mid \mathbf{w}) = -\log p(y \mid \mathbf{x}, \mathbf{w})$.

Computing the gradient:

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Note:

- Computational cost is $O(m)$ $\rightsquigarrow$ increases with the sample size.
- The gradient is an (empirical) expectation.

Stochastic gradient descent

Idea: Approximate the expectation using a small batch of samples $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(B)}$ with B held fixed as sample size grows:

$$\mathbf{g} = \frac{1}{B} \sum_{i=1}^{B} \nabla_{\mathbf{w}} \mathit{loss}(\mathbf{x}^{(i)}, y^{(i)} \mid \mathbf{w}).$$

Updating proceeds using the approximated gradient:

$$\mathbf{w}' = \mathbf{w} - \eta \mathbf{g}.$$

Terminology

- For a batch size of 1 the above approach is referred to as stochastic gradient descents.
- Otherwise, we have mini-batches and the method is referred to as mini-batch gradient descent.

Effect of increasing the batch size

- The standard error of the mean estimated from N examples is

$$SE(\hat{\mu}) = \sqrt{\text{Var}\left(\frac{1}{N}\sum_{i=1}^{N} x_i\right)} = \frac{\sigma}{\sqrt{N}},$$

where σ is the true standard deviation of the samples.

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  - Two estimate of the gradient: (a) based on 100 samples and (b) based on 10000 samples. (b) is 100 times slower, but reduces the standard error of the mean only by a factor 10.

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Observation

$\rightsquigarrow$ SGD also suitable in a streaming setting.

Setting the learning rate

Because the stochastic gradient is noisy, the learning rate η should decrease over time to ensure convergence.

Sufficient conditions

$$\sum_{i=0}^{\infty} \eta_i = \infty \qquad \sum_{i=0}^{\infty} \eta_i^2 < \infty$$

In practice

- Adjust the learning rate during training by, e.g., reducing the learning rate according to a predefined schedule.
- May use different learning rates for different parameters.

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Alternatively: Adaptive learning rates (AdaGrad, Adam, . . .).

Note:

- Different learning rates can be used for different parameters (AdaGrad, . . .).
- For an overview, see `http://ruder.io/optimizing-gradient-descent/`

Momentum

Designed to accelerate learning when faced with high curvature, small or noisy gradients.

Idea

Introduce a variable $\mathbf{v}$ (velocity/momentum) to determine the direction of parameter movement:

$$\mathbf{v} \leftarrow \alpha\mathbf{v} - \eta\nabla_{\mathbf{w}} \left(\frac{1}{N} \sum_{i=1}^{N} \mathit{loss}(\mathbf{x}^{(i)}, y^{(i)} \mid \mathbf{w}) \right)$$

$$\mathbf{w} \leftarrow \mathbf{w} + \mathbf{v}$$

- $\mathbf{v}$ - an exponentially decaying average of the negative gradient

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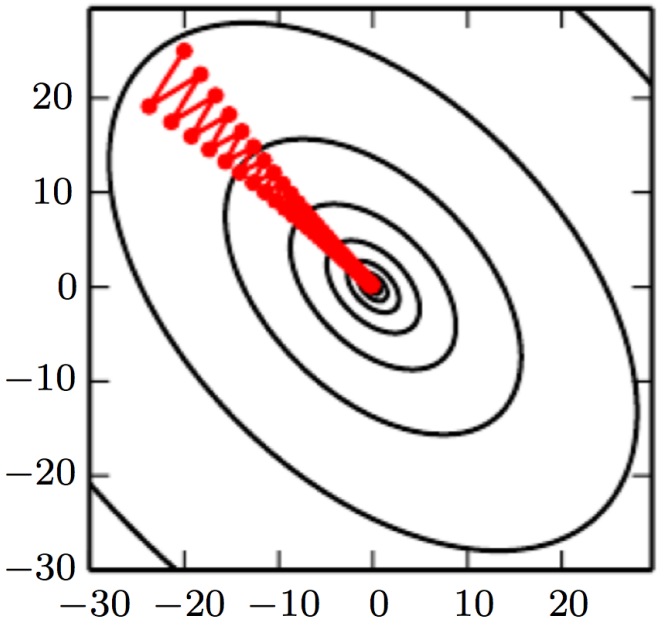
Note: Maximum speed when gradients point in the same direction $-\mathbf{g}$ giving terminal velocity

$$\frac{\eta\|\mathbf{g}\|}{1-\alpha}$$

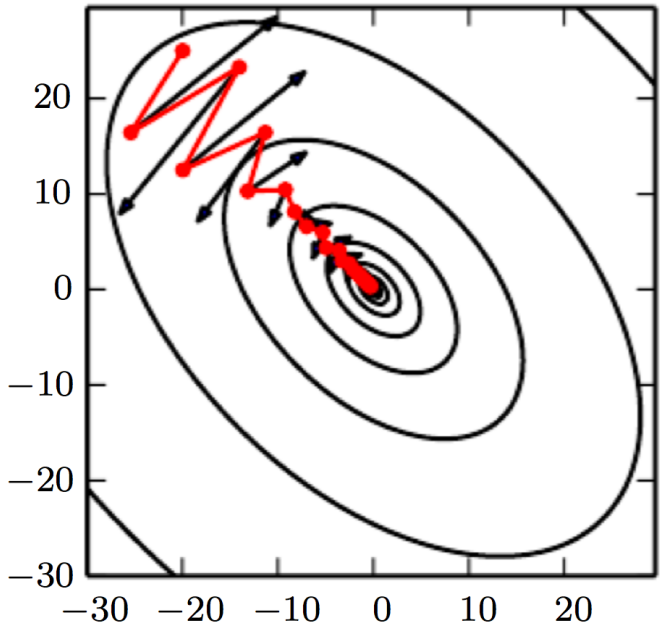
Think in terms of $1/(1-\alpha)$: $\alpha = 0.9 \rightsquigarrow$ multiply max speed by 10 rel. to GDS.

Effect of momentum

Illustration



Without momentum



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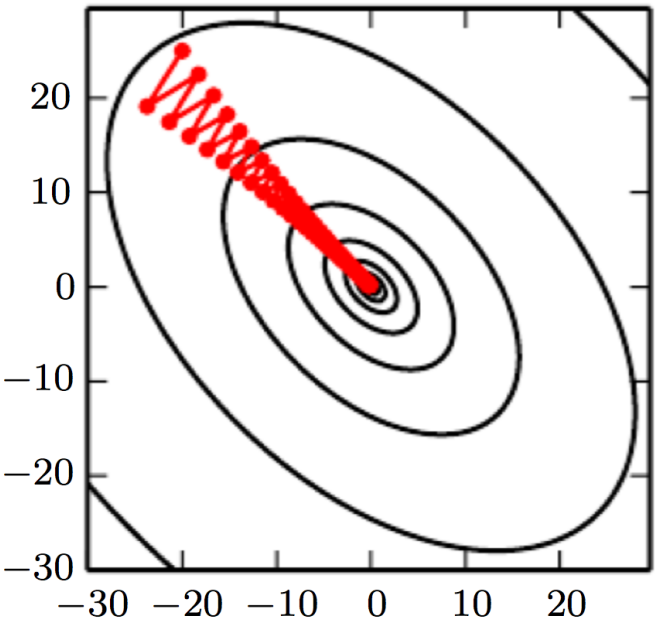

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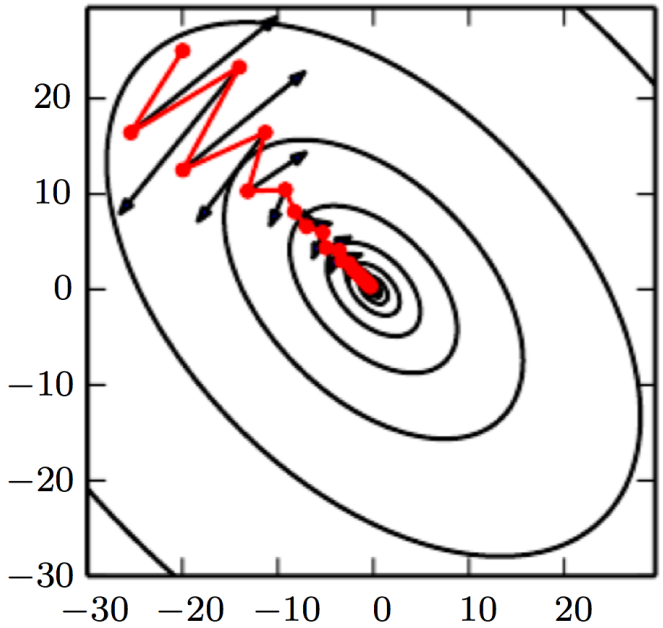


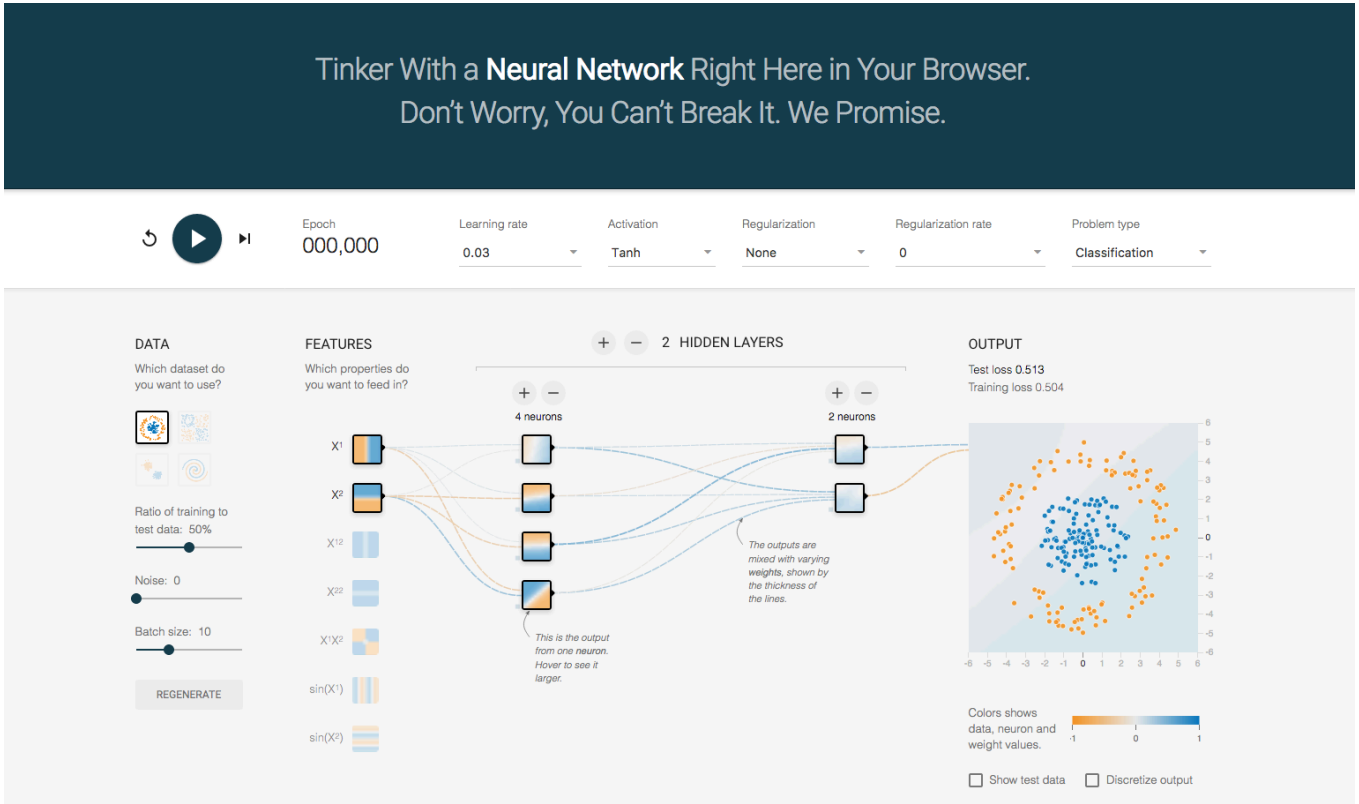
With momentum

Momentum variants: Nesterov momentum

Gradient is evaluated after current velocity has been applied:

$$\mathbf{v} \leftarrow \alpha\mathbf{v} - \eta\nabla_{\mathbf{w}}\left(\frac{1}{N}\sum_{i=1}^{N} \mathit{loss}(\mathbf{x}^{(i)}, y^{(i)} \mid \mathbf{w} + \alpha\mathbf{v})\right)$$

Playground



playground.tensorflow.org